The Wilson Loop Functions on Two Orientations of the Lattice

Abstract

The limits of the renormalised Wilson loop expectations constructed in the paper are covariant under the symmetries of the lattice, and nothing is known about the rotations which are not among them. The smallest instance in which the invariance under such a rotation is a statement about the limits is that of two parallel planar loops whose separation $(5, 0)$ is turned into $(3, 4)$. On a torus whose periods are tilted against the lattice, the reflection which exchanges the two configurations maps the lattice to a second lattice, and we show that the invariance is equivalent to the independence of the limit of the choice between these two lattices. If the blocking factor is a multiple of 5, the construction of the paper can be compared with a run of the renormalisation group which has the same lattices, cubes, and expansions and differs from it in one respect: its fine lattice is turned. We record, in the order of the proofs, the modifications which Bałaban's proofs require for this run. The present text contains the setting, the symmetries which the run retains, its averaging operations with their momentum representation, the description of its blocks as unions of cells with a Poincaré inequality, the operator identities and the L^2-bounds of the first paper on the propagators, and the first part of the paper on the scalar Green's functions. One of Bałaban's methods, the passage to a box with Neumann conditions by reflections, is not available for a turned lattice. The last section lists the proofs which have not been examined.

1 Introduction

The paper constructs, for every ensemble $\mathcal{C}$ of admissible loops on the torus $(\mathbb{R}/T\mathbb{Z})^4$, the limit $W(\mathcal{C})$ of the renormalised expectations $W_K(\mathcal{C})$ as the lattice spacing $\varepsilon_K = uL^{-K}$ tends to zero (Paper, Theorem 1.5), and it shows that the limits are covariant under the translations by vectors of the lattice of the loops and under the reflections in coordinate hyperplanes (Paper, Corollary 1.6). One expects more. The lattice breaks the invariance of the continuum theory under rotations, and the limit in which the lattice is removed should restore it, as far as the torus permits. We ask whether this can be proved for the limits of the paper.

For most ensembles the question cannot be posed within the construction: a rotation which is not a symmetry of the lattice maps a path of the lattice to a path which is not one, and the limits are defined for paths of the lattice only. There is one family of ensembles for which it can. Let the loops lie in planes parallel to e_0 and e_1, and let the rotation turn the plane of e_2 and e_3 and fix the other two coordinates. It then maps each loop to a translate of itself, and it maps the ensemble to an ensemble of the lattice whenever it maps the positions of the planes to points of the lattice. This happens for rotations which are not symmetries of the lattice, because a circle around a point of the square lattice may pass through points which the symmetries of the lattice do not relate: $5^2 = 3^2 + 4^2$. The smallest instance is a loop C with a parallel copy, and the question is whether

$$W(C, C + 5\varepsilon_q e_2) = W(C, C + 3\varepsilon_q e_2 + 4\varepsilon_q e_3) \;, \tag{1.1}$$

where ε_q is the spacing of the lattice of the loops. The two sides are two numbers of one lattice theory, and none of the properties which we know relates them. Nor could these properties relate them: the kernel $(d_2, d_3) \mapsto e^{-|d_2|-|d_3|}$ is continuous, it is invariant under the symmetries of the square lattice, and it is reflection positive in both coordinate directions, since $e^{-(x_2+y_2)}$ is a product and $e^{-|x_3-y_3|}$ is a positive definite kernel, and it takes different values at $(5, 0)$ and $(3, 4)$.

The first difficulty is the torus. The rotation which turns $(5, 0)$ into $(3, 4)$ is not an isometry of $(\mathbb{R}/T\mathbb{Z})^4$, and on this torus the two configurations are not congruent, so that there is no reason for (1.1) to hold there. We therefore change the torus. Writing $z = x_2 + ix_3$, the reflection $\rho\colon z \mapsto \frac{3+4i}{5}\bar{z}$ of the plane maps 5 to $3+4i$, and it is an isometry of the torus whose periods in this plane are $T(2 + i)$ and $iT(2 + i)$, a torus which the lattices of the paper fit as well. On this torus ρ maps the lattice Λ_K of spacing ε_K to a second lattice $\Lambda'_K = \rho\Lambda_K$, tilted against the first, and it maps the lattice theory on the one to the lattice theory on the other. The expectation of the configuration $(5, 0)$ on Λ_K is therefore equal to the expectation of the configuration $(3, 4)$ on Λ'_K, at every K and every coupling (Proposition 4.2). The configuration $(3, 4)$ consists of paths of both lattices, and (1.1) becomes the statement that the limit of its renormalised expectations is the same along the two families of lattices: the independence of the continuum limit of the orientation of the lattice, in the one case in which the two orientations share an observable.

A comparison of two lattice theories is what the paper does when it proves that the limit exists. It compares the construction on Λ_K with that on Λ_{K+1}, two runs of the renormalisation group which share every lattice but the finest (Paper, Section 9.1). The lattices Λ_K and Λ'_K share nothing, but if the blocking factor L is a multiple of 5, then Λ_K is a sublattice of Λ'_{K+1} (Lemma 2.2). A run which starts from the lattice theory on the tilted lattice Λ'_{K+1} and passes, by one renormalisation step, to the lattice Λ_K of the paper, and from there to $\Lambda_{K-1}, \Lambda_{K-2}, \dots$, shares every lattice but the finest with the construction of the paper on Λ_K. Its coarse lattices, its cubes, its localisations, and its expansions are those of the paper. What differs is the fine lattice, on which every background field and every propagator of the run is defined: it is turned by ρ against everything else.

The proofs of Bałaban on which the paper rests are written for a fine lattice of which the coarser lattices are sublattices spanned by multiples of the same basis. Our aim is to record what changes in them when the fine lattice is turned, and only that: each statement which is used is assigned to one of three classes, those which hold with the same proof, those which hold after a modification, which we state and prove, and those which we have not examined. We proceed in the order of the proofs. Sections 2–4 contain the setting: the arithmetic of the two lattices, the torus, and the identity between the expectations which reduces (1.1) to a comparison of two runs. Section 5 determines the symmetries which the tilted run retains; they leave three invariant quadratic forms of the field strength where the symmetries of one lattice leave one, which affects the definition of the running coupling. Section 6 defines the averaging operations of the tilted run and proves the properties which replace those of [2, Sections A and B], and Section 7 gives their momentum representation, in which the turned lattice appears as one fixed trigonometric polynomial multiplying Bałaban's symbols and as a matrix in place of a scalar. Section 8 describes a block as a union of cells of 5^4 points, which gives contours inside the blocks and a Poincaré inequality, and Section 9 proves the L^2-bounds of [2] for the tilted run by counting paths. Section 10 goes through [2] formula by formula, Section 11 through the first two sections of [1], and Section 12 lists what remains.

Throughout, G, L, u, m_T, $T = 2L^{m_T}u$, and $\varepsilon_K = uL^{-K}$ are as in Paper, Section 1.1; in particular L is odd.

2 The Two Lattices

We identify the plane of the coordinates x_2 and x_3 with $\mathbb{C}$ through $z = x_2 + ix_3$, and we write a point of $\mathbb{R}^4$ as (x_0, x_1, z). The points of this plane with integer coordinates are the Gaussian integers $\mathbb{Z}[i] = \mathbb{Z} + i\mathbb{Z}$. We put

$$\omega := \frac{3+4i}{5} = \frac{(2+i)^2}{5} = \frac{2+i}{2-i}\,, \qquad \rho(x_0, x_1, z) := (x_0, x_1, \omega\bar{z})\,. \tag{2.1}$$

Since $|\omega| = 1$ and $\omega\,\overline{\omega\bar{z}} = z$, the map ρ is a linear isometry of $\mathbb{R}^4$ with $\rho^2 = 1$. It fixes e_0 and e_1, it fixes $2 + i$ because $\omega(2 - i) = 2 + i$, and it maps $i(2 + i)$ to $\omega(-i)(2 - i) = -i(2 + i)$; it is therefore the reflection in the hyperplane spanned by e_0, e_1, and $2e_2 + e_3$. We write $e'_\nu := \rho e_\nu$, so that $e'_0 = e_0$, $e'_1 = e_1$, and, because $\omega\bar{1} = \omega$ and $\omega\bar{i} = (4 - 3i)/5$,

$$e'_2 = \tfrac{1}{5}(3e_2 + 4e_3)\,, \quad e'_3 = \tfrac{1}{5}(4e_2 - 3e_3)\,, \qquad e_2 = \tfrac{1}{5}(3e'_2 + 4e'_3)\,, \quad e_3 = \tfrac{1}{5}(4e'_2 - 3e'_3)\,, \tag{2.2}$$

the last two by applying ρ to the first two. In particular $\rho(5e_2) = 3e_2 + 4e_3$.

Lemma 2.1 (Gaussian Integers). *(i) For $w \in \mathbb{Z}[i]$, $\omega w \in \mathbb{Z}[i]$ if and only if $w \in (2-i)\mathbb{Z}[i]$. Consequently $\mathbb{Z}[i] \cap \omega\mathbb{Z}[i] = (2 + i)\mathbb{Z}[i]$.*

(ii) The group $(2 + i)\mathbb{Z}[i]$ has the index 5 in $\mathbb{Z}[i]$ and in $\omega\mathbb{Z}[i]$.

(iii) $\omega\,\overline{(2 + i)\mathbb{Z}[i]} = (2 + i)\mathbb{Z}[i]$.

(iv) $5\omega\mathbb{Z}[i] \subset \mathbb{Z}[i]$ and $5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$, each with the index 25.

Proof. (i) Let $w = a+bi$ with integers a, b. Then $5\omega w = (3a-4b)+(4a+3b)i$, and $\omega w \in \mathbb{Z}[i]$ if and only if $3a \equiv 4b$ and $4a \equiv -3b$ modulo 5. Multiplying the first congruence by 2 and the second by 4, each of them is equivalent to $a \equiv 3b$, that is, to $a + 2b \equiv 0$ modulo 5. On the other hand $(2 - i)(m + ni) = (2m + n) + (2n - m)i$, for which $a + 2b = 5n$; and if $a + 2b = 5n$, then $m := 2n - b$ satisfies $2m + n = 5n - 2b = a$, so that $w = (2 - i)(m + ni)$. This is the first assertion. For such $w = (2 - i)w'$ we have $\omega w = (2 + i)w'$, so that the elements of $\omega\mathbb{Z}[i]$ which lie in $\mathbb{Z}[i]$ are those of $(2 + i)\mathbb{Z}[i]$.

(ii) The multiplication by $2+i$ is the linear map of $\mathbb{Z}^2$ with the columns $(2, 1)$ and $(-1, 2)$, of determinant 5, so that its image has the index 5 in $\mathbb{Z}[i]$. The multiplication by ω^{-1} maps $\omega\mathbb{Z}[i]$ onto $\mathbb{Z}[i]$ and $(2 + i)\mathbb{Z}[i]$ onto $(2 - i)\mathbb{Z}[i]$, which has the index 5 in $\mathbb{Z}[i]$ for the same reason.

(iii) Complex conjugation maps $\mathbb{Z}[i]$ onto itself, so that $\omega\,\overline{(2 + i)\mathbb{Z}[i]} = \omega(2 - i)\mathbb{Z}[i] = (2 + i)\mathbb{Z}[i]$.

(iv) $5\omega = 3 + 4i \in \mathbb{Z}[i]$ gives the first inclusion, and the multiplication by $3 + 4i$ has the determinant 25. Since $\omega\bar{\omega} = 1$ and $5\bar{\omega} = 3 - 4i \in \mathbb{Z}[i]$, we have $5\mathbb{Z}[i] = \omega(5\bar{\omega})\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$, with the index of $(3 - 4i)\mathbb{Z}[i]$ in $\mathbb{Z}[i]$, which is 25. *q.e.d.*

For an integer $K \geq 0$ we consider the two lattices in $\mathbb{R}^4$

$$\Lambda_K := \varepsilon_K\mathbb{Z}^4 = \varepsilon_K(\mathbb{Z}^2 \times \mathbb{Z}[i])\,, \qquad \Lambda'_K := \rho\Lambda_K = \varepsilon_K(\mathbb{Z}^2 \times \omega\mathbb{Z}[i])\,. \tag{2.3}$$

The lattice Λ_K is the lattice of the paper before the periods are divided out, and Λ'_K is a lattice of the same spacing whose axes are the vectors e'_ν.

Lemma 2.2 (The Two Families of Lattices). *(i)* $\Lambda_K \cap \Lambda'_K = \varepsilon_K(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i])$, *a sublattice of the index* 5 *in* Λ_K *and in* Λ'_K, *and* ρ *maps it onto itself.*

(ii) If L *is a multiple of* 5, *then* $\Lambda_K \subset \Lambda'_{K+1}$ *and* $\Lambda'_K \subset \Lambda_{K+1}$, *each with the index* L^4.

Proof. (i) is Lemma 2.1(i)–(iii). (ii) Since $\varepsilon_K = L\varepsilon_{K+1}$ and $L\mathbb{Z}[i] \subset 5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$ by Lemma 2.1(iv), the definition (2.3) gives $\Lambda_K = \varepsilon_{K+1}(L\mathbb{Z}^2 \times L\mathbb{Z}[i]) \subset \Lambda'_{K+1}$. Both are lattices of rank four, and ρ preserves volumes, so that the index is the quotient $\varepsilon_K^4/\varepsilon_{K+1}^4 = L^4$ of the volumes of their cells. Applying ρ gives $\Lambda'_K \subset \Lambda_{K+1}$ with the same index. *q.e.d.*

Thus, for L a multiple of 5, the two families form one chain $\cdots \supset \Lambda'_{K+1} \supset \Lambda_K \supset \Lambda'_{K-1} \supset \Lambda_{K-2} \supset \ldots$, each lattice of the index L^4 in the one before, while no lattice of one family contains a lattice of the same index of the other.

3 One Torus for Both Lattices

The torus $(\mathbb{R}/T\mathbb{Z})^4$ of the paper is of no use for our question: $\rho(Te_2) = \frac{1}{5}T(3e_2 + 4e_3)$ is not a period, so that ρ does not act on it. We need a group of periods which ρ preserves and which lies in both lattices, and Lemma 2.2(i) provides one.

Definition 3.1 (The Tilted Torus). Let

$$\Gamma := T(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i]) \,, \qquad \mathbb{T}_\Gamma := \mathbb{R}^4/\Gamma \,. \tag{3.1}$$

The periods of $\mathbb{T}_\Gamma$ are Te_0, Te_1, and the two vectors $T(2e_2 + e_3)$ and $T(-e_2 + 2e_3)$ of the plane, which are orthogonal and of the length $\sqrt{5}\,T$.

Lemma 3.2 (The Torus and the Lattices). *(i)* $\Gamma \subset \Lambda_K \cap \Lambda'_K$ *for every* $K \geq 0$, *and the quotients* Λ_K/Γ *and* Λ'_K/Γ *have* $5(T/\varepsilon_K)^4$ *elements each.*

(ii) $\rho\Gamma = \Gamma$, *so that* ρ *induces an isometry of* $\mathbb{T}_\Gamma$, *which maps* Λ_K/Γ *onto* Λ'_K/Γ.

(iii) Γ *has the index* 5 *in* $T\mathbb{Z}^4$, *so that* $\mathbb{T}_\Gamma$ *covers the torus of the paper five times.*

Proof. (i) The number $T/\varepsilon_K = 2L^{m_T+K}$ is an integer, so that $\Gamma \subset \varepsilon_K(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i])$, which is $\Lambda_K \cap \Lambda'_K$ by Lemma 2.2(i). The index of Γ in this group is $(T/\varepsilon_K)^4$, and the index of this group in each of the two lattices is 5. (ii) is Lemma 2.1(iii), and $\rho\Lambda_K = \Lambda'_K$ by definition. (iii) is Lemma 2.1(ii). *q.e.d.*

4 The Identity between the Two Lattice Theories

We now define the lattice theory on $\mathbb{T}_\Gamma$ for each of the two lattices and show that ρ carries the one to the other. Let Ξ be Λ_K or Λ'_K. A *bond* of Ξ is an ordered pair $b = (x, y)$ of points of Ξ at the distance ε_K, with the reversed bond $b^{-1} = (y, x)$; the bonds from x are the eight pairs $(x, x \pm \varepsilon_K e_\mu)$ for $\Xi = \Lambda_K$ and $(x, x \pm \varepsilon_K e'_\mu)$ for $\Xi = \Lambda'_K$. A *plaquette* of Ξ is a square of side ε_K with corners in Ξ, and ∂p is its boundary, a closed path of four bonds. The group $\Gamma \subset \Xi$ acts on the bonds and on the plaquettes by translations, and a translation by $\gamma \neq 0$ fixes no bond and no plaquette and reverses no bond. A *gauge field* on (Ξ, Γ) is a map U

from the bonds of Ξ to G with $U(b^{-1}) = U(b)^{-1}$ and $U(b+\gamma) = U(b)$ for every $\gamma \in \Gamma$. The lattice theory at the coupling $g_0 > 0$ is the probability measure

$$\mathrm{d}\mu^{\Xi}_{\Gamma,g_0}(U) \coloneqq \frac{1}{Z^{\Xi}_{\Gamma,g_0}} \exp\Big(-\frac{1}{g_0^2}\sum_p (1 - \operatorname{Re}\operatorname{tr} U(\partial p))\Big)\,\mathrm{d}U\ , \tag{4.1}$$

where the sum runs over one plaquette from each orbit of Γ on the unoriented plaquettes of Ξ and $\mathrm{d}U$ is the product of the normalised Haar measures over one bond from each orbit of Γ on the unoriented bonds. Neither depends on the choices, because U is periodic, $\operatorname{Re}\operatorname{tr}$ is invariant under conjugation and, on unitary matrices, under inversion, and the Haar measure is invariant under inversion. This is the measure of Paper, (1.2), with the group of periods Γ in place of $T\mathbb{Z}^4$ and, for $\Xi = \Lambda'_K$, with the lattice turned by ρ. We write $\mathbb{E}^{\Xi}_{\Gamma,g_0}$ for its expectation. For a closed path C of bonds of Ξ in $\mathbb{R}^4$ and a character χ we put $W_\chi(C;U) \coloneqq \chi(U(C))$, where $U(C)$ is the ordered product of U over the bonds of C, and for a finite family $\mathcal{C} = ((C_1,\chi_1),\dots,(C_n,\chi_n))$ of such paths with characters, $W_{\mathcal{C}} \coloneqq \prod_i W_{\chi_i}(C_i)$. For a linear map O we write $O\mathcal{C}$ for the family of the images OC_i with the same characters.

Lemma 4.1 (Transport)**.** *Let Ξ be Λ_K or Λ'_K, and let O be a linear isometry of $\mathbb{R}^4$ with $O\Gamma = \Gamma$ for which $O\Xi$ is again one of the two lattices. Then, for every family $\mathcal{C}$ of closed paths of bonds of Ξ with characters and every $g_0 > 0$,*

$$\mathbb{E}^{O\Xi}_{\Gamma,g_0}[W_{O\mathcal{C}}] = \mathbb{E}^{\Xi}_{\Gamma,g_0}[W_{\mathcal{C}}]\ . \tag{4.2}$$

Proof. The map O is a bijection from the points of Ξ onto those of $O\Xi$ which preserves distances, so that it maps bonds to bonds, reversed bonds to reversed bonds, and plaquettes to plaquettes with their boundaries; and since $O(b+\gamma) = Ob + O\gamma$ with $O\Gamma = \Gamma$, it maps the orbits of Γ on the bonds and on the plaquettes of Ξ bijectively onto those of $O\Xi$. For a gauge field U on (Ξ,Γ) put $(O_*U)(b) \coloneqq U(O^{-1}b)$ for the bonds b of $O\Xi$. Then O_*U is a gauge field on $(O\Xi,\Gamma)$, the map O_* is a bijection between the two spaces of gauge fields which carries the product of the Haar measures of the one to that of the other, and $(O_*U)(O\gamma) = U(\gamma)$ for every path γ of bonds of Ξ. In particular $(O_*U)(\partial(Op)) = U(\partial p)$ for every plaquette p of Ξ, so that the sums in (4.1) for O_*U on $O\Xi$ and for U on Ξ agree, and O_* carries μ^{Ξ}_{Γ,g_0} to $\mu^{O\Xi}_{\Gamma,g_0}$. Since $W_{O\mathcal{C}}(O_*U) = W_{\mathcal{C}}(U)$, this gives (4.2). *q.e.d.*

We apply the lemma to $O = \rho$ and to the ensembles for which the question of the introduction can be posed. Let $q \geq 0$, and let $C_1,\dots,C_n$ be closed paths of bonds of Λ_q in the plane $\{z = 0\}$, so that all their bonds are parallel to e_0 or e_1. For $d = (d_1,\dots,d_n)$ with $d_j \in \varepsilon_q\mathbb{Z}[i]$ we write

$$\mathcal{C}_d \coloneqq ((C_1+d_1,\chi_1),\dots,(C_n+d_n,\chi_n))\ , \qquad \rho d \coloneqq (\omega\bar{d}_1,\dots,\omega\bar{d}_n)\ , \tag{4.3}$$

where $C_j + d_j$ is the translate of C_j to the plane $\{z = d_j\}$. Since ρ fixes the plane $\{z = 0\}$ pointwise and is linear, $\rho(C_j + d_j) = C_j + \omega\bar{d}_j$, that is, $\rho\mathcal{C}_d = \mathcal{C}_{\rho d}$.

Proposition 4.2 (Reduction)**.** *Let $q \geq 0$, let $C_1,\dots,C_n$ and $\mathcal{C}_d$ be as above, and let $d_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$ for every j. Then $\rho d_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$ for every j, the families $\mathcal{C}_d$ and $\mathcal{C}_{\rho d}$ consist of closed paths of bonds of both Λ_K and Λ'_K for every $K \geq q$, and for every $K \geq q$ and every $g_0 > 0$,*

$$\mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_d}] = \mathbb{E}^{\Lambda'_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}]\ . \tag{4.4}$$

Consequently

$$\mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] - \mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_d}] = \mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] - \mathbb{E}^{\Lambda'_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}]\ . \tag{4.5}$$

Proof. By Lemma 2.1(iii), $\rho d_j = \omega \bar{d}_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$. The sites of $C_j + d_j$ are the points (x_0, x_1, d_j) with $(x_0, x_1, 0)$ a site of C_j; they lie in $\varepsilon_q(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i]) = \Lambda_q \cap \Lambda'_q$, and the bonds of $C_j + d_j$ are parallel to $e_0 = e'_0$ or $e_1 = e'_1$ and of the length ε_q. A straight path of the length ε_q between two points of Λ_q, or of Λ'_q, along one of its axes is a path of L^{K-q} bonds of Λ_K, or of Λ'_K; so $C_j + d_j$ is a closed path of bonds of both lattices, and the same holds for $C_j + \rho d_j$. Lemma 4.1 with $\Xi = \Lambda_K$ and $O = \rho$, for which $O\Gamma = \Gamma$ by Lemma 3.2(ii) and $O\Xi = \Lambda'_K$, gives (4.4) because $\rho \mathcal{C}_d = \mathcal{C}_{\rho d}$, and (4.5) follows by subtracting both sides of (4.4) from $\mathbb{E}^{\Lambda_K}_{\Gamma, g_0}[W_{\mathcal{C}_{\rho d}}]$. *q.e.d.*

The left side of (4.5) measures how far the lattice theory on Λ_K is from being invariant under ρ on this ensemble, and the right side how much the expectation of one ensemble differs between the two lattices. The first instance in which $\mathcal{C}_{\rho d}$ is not the image of $\mathcal{C}_d$ under a symmetry of Λ_q is $n = 2$, $d_1 = 0$, and $d_2 = 5\varepsilon_q = (2+i)(2-i)\varepsilon_q$, for which $\rho d_2 = (3+4i)\varepsilon_q$; this is (1.1). The step factors of the paper do not depend on the position of a loop (Paper, Proposition 7.13), so that $C_j + d_j$ and $C_j + \rho d_j$ have the same normalisation on Λ_K, and (1.1) is the statement that the left side of (4.5), divided by this normalisation, tends to zero as $K \to \infty$ along the couplings of the paper. On the right side, the first term is the object of the construction of the paper on Λ_K, carried out on the torus $\mathbb{T}_\Gamma$, and the second, read at $K+1$, is the object of the tilted run of the introduction.

5 The Symmetries of the Tilted Run

The tilted run has only those symmetries which preserve the lattices of both families. We determine them, and what they leave of the argument by which a single coupling describes the leading term of an effective action.

Lemma 5.1 (The Common Symmetries)**.** *Let $\mathfrak{G}_0$ be the group of the linear isometries O of $\mathbb{R}^4$ with $O\mathbb{Z}^4 = \mathbb{Z}^4$ and $O\rho\mathbb{Z}^4 = \rho\mathbb{Z}^4$. Its elements are the maps*

$$(x_0, x_1, z) \mapsto (A(x_0, x_1), i^k z)\ , \tag{5.1}$$

with A one of the eight signed permutation matrices of the first two coordinates and $k \in \{0,1,2,3\}$. It has 32 elements, it contains the reflections in the hyperplanes $\{x_0 = 0\}$ and $\{x_1 = 0\}$, and it contains no reflection of the plane of x_2 and x_3.

Proof. An isometry O with $O\mathbb{Z}^4 = \mathbb{Z}^4$ permutes the eight vectors $\pm e_\mu$, the elements of $\mathbb{Z}^4$ of the length 1, and an isometry with $O\rho\mathbb{Z}^4 = \rho\mathbb{Z}^4$ permutes the eight vectors $\pm e'_\nu$. By (2.2) the only vectors common to the two sets are $\pm e_0$ and $\pm e_1$. An element O of $\mathfrak{G}_0$ permutes both sets, hence their intersection, so that $Oe_0, Oe_1 \in \{\pm e_0, \pm e_1\}$; the map O therefore preserves the plane of e_0 and e_1 and its orthogonal complement, and it acts on the first as a signed permutation matrix A and on the second as a signed permutation matrix of e_2 and e_3, that is, as $z \mapsto i^k z$ or as $z \mapsto i^k \bar{z}$. Every such map preserves $\mathbb{Z}^4$, and it preserves $\rho\mathbb{Z}^4 = \mathbb{Z}^2 \times \omega\mathbb{Z}[i]$ if and only if its second part preserves $\omega\mathbb{Z}[i]$. The maps $z \mapsto i^k z$ do. The maps $z \mapsto i^k \bar{z}$ send $\omega\mathbb{Z}[i]$ to $\bar{\omega}\mathbb{Z}[i]$, which is equal to $\omega\mathbb{Z}[i]$ only if $\omega^2 \in \mathbb{Z}[i]$, and $\omega^2 = (-7+24i)/25$. This gives (5.1) and the count $8 \cdot 4$. The reflections named are the cases $A = \mathrm{diag}(-1, 1)$ and $A = \mathrm{diag}(1, -1)$ with $k = 0$. *q.e.d.*

The group of all symmetries of $\mathbb{Z}^4$, the signed permutation matrices, has 384 elements and exchanges any two coordinate planes. The group $\mathfrak{G}_0$ does not, and this has the following consequence. Let $F = (F_{\mu\nu})$ be an antisymmetric tensor, on which an isometry O acts by $(OF)_{\mu\nu} = \sum_{\kappa,\lambda} O_{\mu\kappa} O_{\nu\lambda} F_{\kappa\lambda}$.

Lemma 5.2 (Invariant Quadratic Forms)**.** *The real quadratic forms of the antisymmetric tensors F which are invariant under $\mathfrak{G}_0$ are the combinations*

$$\alpha F_{01}^2 + \beta F_{23}^2 + \lambda \sum_{a\in\{0,1\}} \sum_{j\in\{2,3\}} F_{aj}^2 \tag{5.2}$$

with real α, β, λ, a space of dimension three. Those which are invariant under all signed permutation matrices are the multiples of $\sum_{\mu<\nu} F_{\mu\nu}^2$.

Proof. Let q be an invariant quadratic form, a combination of the products of two of the six components F_{01}, F_{23}, F_{aj}. The three elements θ_0, θ_1, and σ of $\mathfrak{G}_0$ which change the sign of x_0, of x_1, and of z multiply each component by a sign: F_{01} by $(-,-,+)$, F_{23} by $(+,+,+)$, F_{0j} by $(-,+,-)$, and F_{1j} by $(+,-,-)$. A product of two components is invariant under the three only if the two have the same signs, so that

$$\mathsf{q}(F) = \alpha F_{01}^2 + \beta F_{23}^2 + \sum_{j,j'} (m^0_{jj'} F_{0j} F_{0j'} + m^1_{jj'} F_{1j} F_{1j'}) \tag{5.3}$$

with symmetric matrices m^0 and m^1. The element of $\mathfrak{G}_0$ which exchanges x_0 and x_1 exchanges F_{0j} and F_{1j}, so that $m^0 = m^1 =: m$. The element $z \mapsto iz$ maps e_2 to e_3 and e_3 to $-e_2$, so that it replaces the vector (F_{a2}, F_{a3}) by $(-F_{a3}, F_{a2})$; the invariance of (5.3) then requires $m_{22} = m_{33}$ and $m_{23} = -m_{23}$, that is, $m = \lambda \mathbf{1}$. This is (5.2). Conversely each of the three forms is invariant: an element (5.1) multiplies F_{01} by $\det A = \pm 1$, fixes F_{23}, because $z \mapsto i^k z$ has the determinant 1, and replaces the matrix (F_{aj}) by its product with A on one side and with a rotation matrix on the other, which preserves the sum of the squares of its entries. Finally a signed permutation matrix which exchanges the planes of two pairs of coordinates exchanges the corresponding components up to signs, so that the invariance under all of them forces $\alpha = \beta = \lambda$. *q.e.d.*

Modification of [8, (1.20)–(1.22), p. 264]. Bałaban defines the coefficient β_{j+1} of the flow of the coupling as a second derivative of the Fourier transform of the vacuum polarisation $\Pi_{j+1,\mu\nu}$ of a step, for one pair $\mu \neq \nu$, and the definition does not depend on the pair because, by his (1.21), the vacuum polarisation is covariant under the rotations which leave the lattice invariant. In the tilted run the covariance holds for the elements of $\mathfrak{G}_0$ only, and by Lemma 5.2 the second derivatives define three numbers, one for each of the forms (5.2): those of the pair $(0,1)$, of the pair $(2,3)$, and of the four pairs (a,j). We do not know whether the three agree, and the consequences for the inductive description of the densities have not been examined.

6 The Averages of the Tilted Run

From here on L is a multiple of 5. After $k \geq 1$ steps of the tilted run, and in the unit of the lattice which it has reached, the situation is that of [2, Section B] with one change: the unit lattice is $\mathbb{Z}^4$ and the lattices of the intermediate steps are $L^{-j}\mathbb{Z}^4$, $1 \leq j \leq k-1$, as there, but the fine lattice is

$$T_\eta \coloneqq \eta\rho\mathbb{Z}^4 = \eta(\mathbb{Z}^2 \times \omega\mathbb{Z}[i]) \,, \qquad \eta = L^{-k} \,, \tag{6.1}$$

in place of $\eta\mathbb{Z}^4$. Its bonds are the pairs $\langle x, x \pm \eta e'_\nu \rangle$. We call a number a a *scale* if $a/(5\eta)$ and $1/a$ are integers; the numbers L^{-j}, $0 \leq j \leq k-1$, are scales, and so is 5η. For a scale a we have $a\mathbb{Z}^4 \subset T_\eta$, because a/η is a multiple of 5 and $5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$ by Lemma 2.1(iv). A real field is a function A on the bonds of T_η with $A(b^{-1}) = -A(b)$, and $A(\gamma) \coloneqq \sum_{b\in\gamma} A(b)$ for a

path γ of bonds. We consider infinite lattices in this section and the next; all the operations commute with the translations by $\mathbb{Z}^4$ and therefore act on the fields which are periodic with respect to a subgroup of $\mathbb{Z}^4$.

Bałaban's averages are sums, over the points x of a cube of the fine lattice, of the values of a function at x or of the sums of a field over the straight segment $[x, x(b)]$ from x to its translate by a bond b of the coarse lattice [2, (1.6), (1.11), (1.18), pp. 18–20]. In the tilted run the cubes keep their meaning, as the sets of the points of T_η in a cube of the coarse lattice, but for the directions e_2 and e_3 the segment does not consist of bonds of T_η, and we replace it by a path which follows it.

Definition 6.1 (Blocks, Staircases, and Averages)**.** Let a be a scale.

(i) The *block* of $y \in a\mathbb{Z}^4$ is $B_a(y) \coloneqq \{x \in T_\eta \,|\, y_\mu \le x_\mu < y_\mu + a,\ \mu = 0, \dots, 3\}$.

(ii) For $x \in T_\eta$ the *staircases* from x are the following paths of bonds of T_η. For $\mu \in \{0, 1\}$, $\sigma^a_\mu(x)$ is the straight path of a/η bonds in the direction e_μ. The path $\sigma^a_2(x)$ repeats $a/(5\eta)$ times the seven steps $\eta e'_3, \eta e'_2, \eta e'_3, \eta e'_2, \eta e'_3, \eta e'_2, \eta e'_3$, and $\sigma^a_3(x)$ repeats $a/(5\eta)$ times the seven steps $\eta e'_2, -\eta e'_3, \eta e'_2, -\eta e'_3, \eta e'_2, -\eta e'_3, \eta e'_2$.

(iii) For a function λ on T_η and $y \in a\mathbb{Z}^4$,

$$(Q'_a\lambda)(y) \coloneqq \left(\frac{\eta}{a}\right)^4 \sum_{x \in B_a(y)} \lambda(x)\ . \tag{6.2}$$

(iv) For a real field A and a bond $b = \langle y, y + ae_\mu\rangle$ of $a\mathbb{Z}^4$,

$$(Q_a A)(b) \coloneqq \left(\frac{\eta}{a}\right)^5 \sum_{x \in B_a(y)} A(\sigma^a_\mu(x))\ , \tag{6.3}$$

and $(Q_a A)(b^{-1}) \coloneqq -(Q_a A)(b)$.

We write $Q'_k \coloneqq Q'_1$ and $Q_k \coloneqq Q_1$.

For $a = 1$ these are the operators Q'_k of [2, p. 20] and Q_k of [2, (1.18), p. 20], with T_η turned and with $\sigma^1_\mu(x)$ in place of $[x, x(b)]$, and for the directions e_0 and e_1 nothing has changed. For two scales a and a' with a'/a an integer we also need Bałaban's own averages between the lattices $a\mathbb{Z}^4$ and $a'\mathbb{Z}^4$, which we denote by

$$(Q'^{\mathrm{st}}_{a,a'}\lambda)(y) \coloneqq \left(\frac{a}{a'}\right)^4 \sum_z \lambda(z)\ , \qquad (Q^{\mathrm{st}}_{a,a'}B)(\langle y, y+a'e_\mu\rangle) \coloneqq \left(\frac{a}{a'}\right)^5 \sum_z B([z, z+a'e_\mu])\ , \tag{6.4}$$

the sums over the points $z \in a\mathbb{Z}^4$ with $y_\mu \le z_\mu < y_\mu + a'$ for every μ, for functions λ and fields B on $a\mathbb{Z}^4$.

Lemma 6.2 (Blocks and Staircases)**.** *Let a and a' be scales with a'/a an integer.*

(i) The blocks $B_a(y)$, $y \in a\mathbb{Z}^4$, are pairwise disjoint, their union is T_η, each has $(a/\eta)^4$ points, and $B_a(y + w) = B_a(y) + w$ for $w \in a\mathbb{Z}^4$. The block $B_{a'}(y)$ is the union of the blocks $B_a(z)$ over the points $z \in a\mathbb{Z}^4$ with $y_\mu \le z_\mu < y_\mu + a'$.

(ii) The staircase $\sigma^a_\mu(x)$ ends at $x + ae_\mu$, and $\sigma^{a'}_\mu(x)$ is the concatenation of the staircases $\sigma^a_\mu(x + nae_\mu)$, $0 \le n < a'/a$. Each site of $\sigma^a_\mu(x)$ is at a distance of at most $\frac{3}{5}\eta$ from the segment $[x, x + ae_\mu]$.

Proof. (i) Since $a\mathbb{Z}^4$ is a subgroup of T_η and the cube $y + [0,a)^4$ contains exactly one point of every class of $\mathbb{R}^4$ modulo $a\mathbb{Z}^4$, the block $B_a(y)$ contains exactly one point of every class of T_η modulo $a\mathbb{Z}^4$. The number of these classes is the quotient a^4/η^4 of the volumes of the cells of the two lattices, because ρ preserves volumes. The remaining assertions follow from the corresponding ones for the cubes.

(ii) For $\mu \in \{0,1\}$ there is nothing to prove. By (2.2), seven steps of σ_2^a add up to $\eta(3e_2' + 4e_3') = 5\eta e_2$ and seven steps of σ_3^a to $\eta(4e_2' - 3e_3') = 5\eta e_3$, which gives the end point and the concatenation. The distance of a point $\eta(ne_2' + n'e_3')$ from the line through the origin in the direction $3e_2' + 4e_3'$ is $\eta|4n - 3n'|/5$. After one to seven steps of σ_2^a from the origin the pair (n,n') is $(0,1)$, $(1,1)$, $(1,2)$, $(2,2)$, $(2,3)$, $(3,3)$, $(3,4)$, with $|4n-3n'| = 3,1,2,2,1,3,0$, and the foot of the perpendicular lies between the origin and $5\eta e_2$ in each case; the eighth site lies on the line, so that the same holds in every period. For σ_3^a the pairs are $(1,0)$, $(1,-1)$, $(2,-1)$, $(2,-2)$, $(3,-2)$, $(3,-3)$, $(4,-3)$, and the distance from the line in the direction $4e_2' - 3e_3'$ is $\eta|3n + 4n'|/5$, with the same values. *q.e.d.*

The next lemma replaces the calculation [2, (1.16)–(1.18), p. 20] by which a composition of averages is an average, and it contains a second statement which has no counterpart there: the tilted average is one of Bałaban's averages, preceded by an average over cells of 5^4 points which is the same at every step.

Lemma 6.3 (Composition)**.** *For scales a and a' with a'/a an integer,*

$$Q_{a'} = Q^{\mathrm{st}}_{a,a'} Q_a \ , \qquad Q'_{a'} = Q'^{\mathrm{st}}_{a,a'} Q'_a \ . \tag{6.5}$$

In particular $Q_k = Q^{\mathrm{st}}_{L^{-1},1} Q_{L^{-1}}$ and $Q_k = Q^{\mathrm{st}}_{5\eta,1} Q_{5\eta}$, and likewise for Q'_k.

Proof. Let $b = \langle y, y + a'e_\mu \rangle$, and let z run over the points of $a\mathbb{Z}^4$ with $y_\nu \le z_\nu < y_\nu + a'$. The segment $[z, z + a'e_\mu]$ consists of the a'/a bonds $\langle z + nae_\mu, z + (n+1)ae_\mu \rangle$ of $a\mathbb{Z}^4$, so that by (6.4) and (6.3)

$$\begin{aligned} (Q^{\mathrm{st}}_{a,a'} Q_a A)(b) &= \Big(\frac{a}{a'}\Big)^5 \Big(\frac{\eta}{a}\Big)^5 \sum_z \sum_{n=0}^{a'/a-1} \sum_{x \in B_a(z+nae_\mu)} A(\sigma_\mu^a(x)) \\ &= \Big(\frac{\eta}{a'}\Big)^5 \sum_z \sum_{x \in B_a(z)} \sum_{n=0}^{a'/a-1} A(\sigma_\mu^a(x + nae_\mu)) \ , \end{aligned} \tag{6.6}$$

where we have used $B_a(z + nae_\mu) = B_a(z) + nae_\mu$. By Lemma 6.2(ii) the inner sum is $A(\sigma_\mu^{a'}(x))$, and by Lemma 6.2(i) the points x run over $B_{a'}(y)$; this is $(Q_{a'}A)(b)$. The proof for Q' is the last assertion of Lemma 6.2(i). *q.e.d.*

Lemma 6.4 (The Identities of the Averages)**.** *Let a be a scale, and write $(\partial^\eta \lambda)(b) := \eta^{-1}(\lambda(b_+) - \lambda(b_-))$ for a function λ on T_η and a bond $b = \langle b_-, b_+ \rangle$ of T_η, and ∂^a for the same operation on $a\mathbb{Z}^4$.*

(i) $Q_a \partial^\eta = \partial^a Q'_a$.

(ii) Q'_a maps the constant function 1 to 1, and so does its adjoint. For $v \in \mathbb{R}^4$, Q_a maps the constant field $A(\langle x, x + \eta e'_\nu \rangle) := v \cdot e'_\nu$ on T_η to the constant field $B(\langle y, y + ae_\mu \rangle) := v \cdot e_\mu$ on $a\mathbb{Z}^4$, and its adjoint maps B to A. Here the adjoints are taken with respect to the sums over the points, or the positive bonds, with the weights η^4 on T_η and a^4 on $a\mathbb{Z}^4$.

(iii) Q_a and Q'_a commute with the translations by the vectors of $a\mathbb{Z}^4$.

Proof. (i) The sum of $\partial^\eta\lambda$ over a path from x to x' is $\eta^{-1}(\lambda(x')-\lambda(x))$, so that $(Q_a\partial^\eta\lambda)(\langle y, y+ae_\mu\rangle) = (\eta/a)^5\eta^{-1}\sum_{x\in B_a(y)}(\lambda(x+ae_\mu)-\lambda(x))$ by Lemma 6.2(ii). Since $x\mapsto x+ae_\mu$ maps $B_a(y)$ onto $B_a(y+ae_\mu)$, this is $a^{-1}((Q'_a\lambda)(y+ae_\mu)-(Q'_a\lambda)(y))$.

(ii) $Q'_a1 = 1$ is the count of Lemma 6.2(i), and $(Q'^*_a\omega)(x) = \omega(y)$ for $x\in B_a(y)$, because every point lies in one block. The sum of the constant field A over a path from x to x' is $\eta^{-1}v\cdot(x'-x)$, so that every term of (6.3) is $\eta^{-1}av\cdot e_\mu$, and $(\eta/a)^5(a/\eta)^4\eta^{-1}a = 1$. For the adjoint we count the staircases through a bond. By (6.3) and the weights, $(Q^*_aB)(b') = (\eta/a)\sum_\mu\sum_x \pm B(\langle y(x), y(x)+ae_\mu\rangle)$ for a positive bond b' of T_η, the inner sum over the points $x\in T_\eta$ whose staircase $\sigma^a_\mu(x)$ contains b' or its reverse, with the sign $-$ in the second case, and with $y(x)$ the corner of the block of x. A bond in the direction e_0 or e_1 lies on a/η staircases of that direction and on no other. Each period of σ^a_2 has three steps $\eta e'_2$ and four steps $\eta e'_3$, and each period of σ^a_3 has four steps $\eta e'_2$ and three steps $-\eta e'_3$; since every translate of a staircase by a vector of T_η is a staircase, a bond in the direction e'_2 lies on $3a/(5\eta)$ staircases σ^a_2 and on $4a/(5\eta)$ staircases σ^a_3, and a bond in the direction e'_3 on $4a/(5\eta)$ staircases σ^a_2 and, reversed, on $3a/(5\eta)$ staircases σ^a_3. For the constant field B this gives the values $\frac15(3v\cdot e_2+4v\cdot e_3) = v\cdot e'_2$ and $\frac15(4v\cdot e_2-3v\cdot e_3) = v\cdot e'_3$ by (2.2).

(iii) The blocks and the staircases are carried to each other by these translations. *q.e.d.*

Statement (i) is [2, (1.55), p. 27], and with it [2, (1.9), (1.13), (1.20)]. Statement (ii) says that the average of a constant field is the same constant vector, with its components in the basis e'_ν replaced by those in the basis e_μ. On the constant fields Q_a is therefore an isometry with $Q^*_aQ_a = 1$, where Bałaban's average is the identity, and Q_a and Q'_a map the orthogonal complements of the constants to the orthogonal complements of the constants.

7 The Momentum Representation

Bałaban's bounds on the operators built from Q_k, Q'_k, and the Laplace operator of the fine lattice are read off from their momentum representation, in which Q'_k is the multiplication by the function u_k of [2, (1.31), p. 23] and Q_k the multiplication of the component μ by u_kv_μ [2, (1.61), p. 28]. Both are products of quotients of sines, and their properties are evident from this form. For the tilted averages the factorisation $Q_k = Q^{\mathrm{st}}_{5\eta,1}Q_{5\eta}$ of Lemma 6.3 gives the corresponding form: Bałaban's functions for the lattice of the spacing 5η, multiplied by one trigonometric polynomial of 25 terms for the cell, and with a matrix of polynomials of at most four terms in place of the scalar v_μ in the two turned directions.

For $p\in\mathbb{R}^4$ we write $e_p(x) := e^{ip\cdot x}$, and for a number $a>0$

$$\begin{aligned} \partial^a_\mu(p) &:= \frac{e^{iap_\mu}-1}{a}\,, \qquad & \partial'_\nu(p) &:= \frac{e^{i\eta p\cdot e'_\nu}-1}{\eta}\,, \\ \Delta(p) &:= \sum_\nu|\partial'_\nu(p)|^2\,, \qquad & \Delta_0(p) &:= \sum_\mu|\partial^1_\mu(p)|^2\,. \end{aligned} \tag{7.1}$$

The functions ∂'_ν and Δ are the symbols of the difference operators and of the Laplace operator of T_η, and ∂^1_μ and Δ_0 those of the unit lattice, as in [2, (1.31), p. 23, and p. 24]. For numbers a and a' with a'/a an integer we put

$$D^{a,a'}_\mu(p) := \frac{a}{a'}\sum_{n=0}^{a'/a-1} e^{ianp_\mu}\,, \tag{7.2}$$

which is $\partial_\mu^{a'}(p)/\partial_\mu^a(p)$ wherever $\partial_\mu^a(p) \neq 0$ and satisfies $D_\mu^{a,a''} = D_\mu^{a,a'} D_\mu^{a',a''}$. Bałaban's functions are $u_k = \prod_\mu D_\mu^{\eta,1}$ and $v_\mu = D_\mu^{\eta,1}$. The cell enters through the set $\mathsf{C} \coloneqq \{c \in \omega\mathbb{Z}[i] \,|\, 0 \leq \operatorname{Re} c < 5,\ 0 \leq \operatorname{Im} c < 5\}$, which has 25 elements by Lemma 6.2(i) for $a = 5\eta$, and through the functions of $\theta = (\theta_2, \theta_3)$

$$\begin{aligned} \kappa(\theta) &\coloneqq \frac{1}{25} \sum_{c \in \mathsf{C}} e^{i\theta \cdot c} \,, \\ s_{22}(\theta) &\coloneqq e^{i\theta \cdot e_3'} + e^{i\theta \cdot (e_2' + 2e_3')} + e^{i\theta \cdot (2e_2' + 3e_3')} \,, \\ s_{23}(\theta) &\coloneqq 1 + e^{i\theta \cdot (e_2' + e_3')} + e^{i\theta \cdot (2e_2' + 2e_3')} + e^{i\theta \cdot (3e_2' + 3e_3')} \,, \\ s_{32}(\theta) &\coloneqq 1 + e^{i\theta \cdot (e_2' - e_3')} + e^{i\theta \cdot (2e_2' - 2e_3')} + e^{i\theta \cdot (3e_2' - 3e_3')} \,, \\ s_{33}(\theta) &\coloneqq -e^{i\theta \cdot (e_2' - e_3')} - e^{i\theta \cdot (2e_2' - 2e_3')} - e^{i\theta \cdot (3e_2' - 3e_3')} \,, \end{aligned} \tag{7.3}$$

where c is read as the vector $(\operatorname{Re} c, \operatorname{Im} c)$ and $\theta \cdot e_\nu'$ is the product with the components of e_ν' in the plane.

Lemma 7.1 (The Symbols of the Tilted Averages)**.** *Let $p \in \mathbb{R}^4$.*

(i) $Q_k' e_p = u_k(p)\, e_p$ on $\mathbb{Z}^4$, with

$$\begin{aligned} u_k(p) &\coloneqq \sum_{x \in B_1(0)} \eta^4 e^{ip \cdot x} \\ &= D_0^{\eta,1}(p)\, D_1^{\eta,1}(p)\, D_2^{5\eta,1}(p)\, D_3^{5\eta,1}(p)\, \kappa(\eta p_2, \eta p_3) \,. \end{aligned} \tag{7.4}$$

(ii) For $w = (w_\nu) \in \mathbb{C}^4$ and the field $A(\langle x, x + \eta e_\nu' \rangle) \coloneqq w_\nu e^{ip \cdot x}$, we have $(Q_k A)(\langle y, y + e_\mu \rangle) = u_k(p) \sum_\nu V_{\mu\nu}(p) w_\nu\, e^{ip \cdot y}$, where $V_{\mu\mu}(p) \coloneqq D_\mu^{\eta,1}(p)$ for $\mu \in \{0, 1\}$,

$$V_{\mu\nu}(p) \coloneqq \tfrac{1}{5} D_\mu^{5\eta,1}(p)\, s_{\mu\nu}(\eta p_2, \eta p_3) \tag{7.5}$$

for $\mu, \nu \in \{2, 3\}$, and $V_{\mu\nu} \coloneqq 0$ for the other pairs.

(iii) $\sum_\nu V_{\mu\nu}(p)\, \partial_\nu'(p) = \partial_\mu^1(p)$ for every μ.

(iv) u_k and V are periodic with respect to $2\pi\eta^{-1}\rho\mathbb{Z}^4$, which is a subgroup of $2\pi\mathbb{Z}^4$; $u_k(0) = 1$; $u_k(l) = 0$ for every $l \in 2\pi\mathbb{Z}^4$ which does not lie in $2\pi\eta^{-1}\rho\mathbb{Z}^4$; and $V_{\mu\nu}(0) = e_\mu \cdot e_\nu'$.

Proof. (i) By (6.2), $(Q_k' e_p)(y) = \sum_{x \in B_1(0)} \eta^4 e^{ip \cdot (y+x)}$, which is the first equality. By Lemma 6.2(i), $B_1(0)$ is the disjoint union of the sets $z + B_{5\eta}(0)$ over the $z \in 5\eta\mathbb{Z}^4$ with $0 \leq z_\mu < 1$, and $B_{5\eta}(0)$ is the set of the points $\eta(n_0, n_1, c)$ with $n_0, n_1 \in \{0, \dots, 4\}$ and $c \in \mathsf{C}$. Hence

$$\begin{aligned} u_k(p) &= \Big(\sum_z (5\eta)^4 e^{ip \cdot z} \Big) \Big(\frac{1}{5^4} \sum_{n_0, n_1 = 0}^{4} \sum_{c \in \mathsf{C}} e^{i\eta(n_0 p_0 + n_1 p_1)} e^{i\eta(p_2, p_3) \cdot c} \Big) \\ &= \prod_{\mu=0}^{3} D_\mu^{5\eta,1}(p) \cdot D_0^{\eta,5\eta}(p)\, D_1^{\eta,5\eta}(p)\, \kappa(\eta p_2, \eta p_3) \,, \end{aligned} \tag{7.6}$$

and $D_\mu^{\eta,5\eta} D_\mu^{5\eta,1} = D_\mu^{\eta,1}$.

(ii) For $\mu \in \{0, 1\}$ the sum of A over $\sigma_\mu^1(x)$ is $w_\mu e^{ip \cdot x} \sum_{n < 1/\eta} e^{i\eta n p_\mu} = \eta^{-1} D_\mu^{\eta,1}(p) w_\mu e^{ip \cdot x}$, and (6.3) with (i) gives the assertion. For $\mu = 2$, the bonds of the first period of $\sigma_2^1(x)$ start at

the points $x+\eta c$ with $c = 0$, e'_3, $e'_2+e'_3$, $e'_2+2e'_3$, $2e'_2+2e'_3$, $2e'_2+3e'_3$, $3e'_2+3e'_3$, and they have the directions $e'_3, e'_2, e'_3, e'_2, e'_3, e'_2, e'_3$; the sum of A over them is $e^{ip\cdot x}(s_{22}(\eta p)w_2 + s_{23}(\eta p)w_3)$. The n-th period is the translate of the first by $5\eta n e_2$, so that the sum of A over $\sigma_2^1(x)$ is this number multiplied by $\sum_{n<1/(5\eta)} e^{5i\eta n p_2} = (5\eta)^{-1} D_2^{5\eta,1}(p)$. For $\mu = 3$ the steps in the direction e'_2 start at the points $x + \eta j(e'_2 - e'_3)$, $j = 0, \dots, 3$, and a step $-\eta e'_3$ from $x+\eta c$ is the reversed bond $\langle x+\eta(c-e'_3), x+\eta c\rangle$, on which A takes the value $-w_3 e^{ip\cdot(x+\eta(c-e'_3))}$; the three such steps end at $x + \eta j(e'_2 - e'_3)$, $j = 1, 2, 3$. This gives s_{32} and s_{33}.

(iii) Multiply each term of $s_{\mu\nu}(\theta)$ by $e^{i\theta\cdot e'_\nu} - 1$ and add: each step of a period from c to c' contributes $e^{i\theta\cdot c'} - e^{i\theta\cdot c}$, a step $-\eta e'_3$ included, since $-(e^{i\theta\cdot e'_3} - 1)e^{i\theta\cdot(c-e'_3)} = e^{i\theta\cdot(c-e'_3)} - e^{i\theta\cdot c}$. The sum over a period telescopes to $e^{5i\theta_\mu} - 1$, because the period ends at $5e_\mu$. Hence $\sum_\nu V_{\mu\nu}\partial'_\nu = \frac{1}{5} D_\mu^{5\eta,1}(p)\eta^{-1}(e^{5i\eta p_\mu} - 1) = D_\mu^{5\eta,1}(p)\,\partial_\mu^{5\eta}(p) = \partial_\mu^1(p)$ for $\mu \in \{2,3\}$, and for $\mu \in \{0,1\}$ it is $D_\mu^{\eta,1}\partial_\mu^\eta = \partial_\mu^1$.

(iv) For $n \in \mathbb{Z}^4$ and $x \in T_\eta$ the product $2\pi\eta^{-1}\rho n \cdot x$ is a multiple of 2π, because ρ is an isometry and $\rho x \in \eta\mathbb{Z}^4$; this gives the periodicity of u_k, and that of V follows in the same way from the positions of the bonds. The inclusion holds because $\eta^{-1}\omega\mathbb{Z}[i] \subset \mathbb{Z}[i]$ by Lemma 2.1(iv). The value $u_k(0) = 1$ is the count of Lemma 6.2(i). For $l \in 2\pi\mathbb{Z}^4$ the function $x \mapsto e^{il\cdot x}$ is a character of the group T_η which is trivial on $\mathbb{Z}^4$, and $B_1(0)$ contains one point of every class of T_η modulo $\mathbb{Z}^4$; the sum of a nontrivial character over a finite group vanishes, and the character is trivial only if $l\cdot x \in 2\pi\mathbb{Z}$ for every $x \in T_\eta$, that is, $l \in 2\pi\eta^{-1}\rho\mathbb{Z}^4$. Finally $s_{22}(0) = 3$, $s_{23}(0) = 4$, $s_{32}(0) = 4$, $s_{33}(0) = -3$, which are the numbers $5e_\mu\cdot e'_\nu$ of (2.2). *q.e.d.*

Statement (iii) is the identity $Q_k\partial = \partial_1 Q'_k$ in the momentum representation, and (iv) contains the two properties of u_k which Bałaban uses after [2, (1.33), p. 23]. In his setting a momentum p of the fine lattice is written as $p = p' + l$ with p' in the dual of the unit lattice and $l \in 2\pi\mathbb{Z}^4$, the classes of l being taken modulo $2\pi\eta^{-1}\mathbb{Z}^4$. Here they are taken modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$, and the number of the classes is again η^{-4}.

The bounds on u_k are those of [2, (1.36), p. 24] with one change. For $p \in \mathbb{R}^4$ let q_μ, for $\mu \in \{0,1\}$, be the number with $q_\mu \equiv p_\mu$ modulo $2\pi/\eta$ and $-\pi/\eta < q_\mu \le \pi/\eta$, and for $\mu \in \{2,3\}$ the number with $q_\mu \equiv p_\mu$ modulo $2\pi/(5\eta)$ and $-\pi/(5\eta) < q_\mu \le \pi/(5\eta)$, so that $(p_2, p_3) = (q_2, q_3) + \frac{2\pi}{5\eta} m$ with $m \in \mathbb{Z}[i]$. We call p *central* if $m \in (3+4i)\mathbb{Z}[i]$.

Lemma 7.2 (Bounds on the Symbols). *(i)* $\kappa(0) = 1$; $\kappa(\theta + \frac{2\pi}{5}m) = \kappa(\theta)$ *for* $m \in (3+4i)\mathbb{Z}[i]$; $\kappa(\frac{2\pi}{5}m) = 0$ *for* $m \in \mathbb{Z}[i] \setminus (3+4i)\mathbb{Z}[i]$; *and* $|\kappa(\theta) - \kappa(\theta')| \le 5\sqrt{2}|\theta - \theta'|$.

(ii) For every $p \in \mathbb{R}^4$, *with* q *as above,*

$$|u_k(p)| \le \prod_{\mu=0}^{3} \min\Big(1, \frac{\pi|\partial_\mu^1(p)|}{2|q_\mu|}\Big) \cdot \begin{cases} 1 & \text{if } p \text{ is central,} \\ \min(1, 5\sqrt{2}\,\eta|(q_2, q_3)|) & \text{otherwise,} \end{cases} \tag{7.7}$$

and $|V_{\mu\nu}(p)| \le \frac{4}{5}\min(1, \pi|\partial_\mu^1(p)|/(2|q_\mu|))$ *for* $\mu, \nu \in \{2,3\}$; *a quotient with* $q_\mu = 0$ *is read as* ∞.

(iii) For $p \in [-\pi, \pi]^4$,

$$|u_k(p)| \ge \Big(\frac{2}{\pi}\Big)^4 (1 - 10\pi\eta)\,, \tag{7.8}$$

and the matrix $V^{\mathrm{pl}}(p) := (V_{\mu\nu}(p))_{\mu,\nu\in\{2,3\}}$ *satisfies* $|V^{\mathrm{pl}}(p)w| \ge \frac{2}{\pi}(1 - 32\eta)|w|$ *for every* $w \in \mathbb{C}^2$.

Proof. (i) $\kappa(0) = 1$ because C has 25 elements. For $m \in \mathbb{Z}[i]$ the function $c \mapsto e^{i\frac{2\pi}{5} m\cdot c}$ is a character of the group $\omega\mathbb{Z}[i]$, trivial on $5\mathbb{Z}[i]$, since $m \cdot c \in 5\mathbb{Z}$ for $c \in 5\mathbb{Z}[i]$; and C contains one element of every class of $\omega\mathbb{Z}[i]$ modulo $5\mathbb{Z}[i]$. The character is trivial on $\omega\mathbb{Z}[i]$ if and only if $m \cdot \omega w \in 5\mathbb{Z}$ for every $w \in \mathbb{Z}[i]$, that is, since $m \cdot \omega w = \mathrm{Re}(\bar\omega m \bar w)$, if and only if $\bar\omega m \in 5\mathbb{Z}[i]$, which means $m \in 5\omega\mathbb{Z}[i] = (3+4i)\mathbb{Z}[i]$. If it is trivial, multiplying the terms of κ by it changes nothing, which is the periodicity; if it is not, its sum over the classes vanishes. The last assertion holds because $|c| < 5\sqrt{2}$ for $c \in \mathsf{C}$.

(ii) For a number a with $1/a$ an integer and a real t with $|at| \le \pi$ we have $|e^{iat} - 1|/a = 2|\sin(at/2)|/a \ge 2|t|/\pi$, so that $|D^{a,1}_\mu(p)| = |\partial^1_\mu(p)|/|\partial^a_\mu(q)| \le \pi|\partial^1_\mu(p)|/(2|q_\mu|)$ for $a = \eta$, $\mu \in \{0,1\}$, and for $a = 5\eta$, $\mu \in \{2,3\}$, because ∂^a_μ has the period $2\pi/a$ in p_μ; and $|D^{a,1}_\mu| \le 1$ by (7.2). Further $|\kappa| \le 1$, and if p is not central, then $\kappa(\eta(p_2,p_3)) = \kappa(\eta(q_2,q_3)+\frac{2\pi}{5}m) - \kappa(\frac{2\pi}{5}m)$ by (i), which is at most $5\sqrt{2}\,\eta|(q_2,q_3)|$ in modulus. With (7.4) this is (7.7), and the bound on V follows from (7.5) and $|s_{\mu\nu}| \le 4$.

(iii) For $|t| \le \pi$ and $a \le 1$ we have $|e^{it} - 1| = 2|\sin(t/2)| \ge 2|t|/\pi$ and $|e^{iat} - 1|/a \le |t|$, so that $|D^{a,1}_\mu(p)| \ge 2/\pi$; and $|\kappa(\eta(p_2,p_3)) - 1| \le 5\sqrt{2}\,\eta|(p_2,p_3)| \le 10\pi\eta$ by (i). This gives (7.8). Let R_0 be the matrix with the entries $\frac15 s_{\mu\nu}(0) = e_\mu \cdot e'_\nu$, which is orthogonal by (2.2). Each $s_{\mu\nu}$ has at most four terms $\pm e^{i\theta\cdot c}$ with $|c| \le 3\sqrt{2}$, so that $|\frac15 s_{\mu\nu}(\theta) - \frac15 s_{\mu\nu}(0)| \le \frac{12\sqrt{2}}{5}|\theta|$, and the norm of the matrix of these differences is at most twice this number. For $\theta = \eta(p_2,p_3)$, $|\theta| \le \sqrt{2}\,\pi\eta$, this is at most $\frac{48\pi}{5}\eta \le 32\eta$. Since $V^{\mathrm{pl}}(p)$ is the product of the diagonal matrix of the numbers $D^{5\eta,1}_\mu(p)$, of modulus at least $2/\pi$, with the matrix $\frac15 s(\theta)$, the assertion follows. *q.e.d.*

The first product in (7.7) is Bałaban's bound $\prod_\mu |p'_\mu|/|p'_\mu + l_\mu|$, with the difference that in the turned directions q_μ is the representative of p_μ in an interval of the length $2\pi/(5\eta)$ rather than $2\pi/\eta$. Each value of q therefore belongs to 25 of the classes of l; one of them is central, and on the other 24 the symbol carries the additional factor $\eta|q|$, while $\Delta(p)$ is of the order of η^{-2} there.

As a first application we give the bound on the operator $\phi = Q_k \Delta^{-1} Q_k^*$ of [2, (1.58), (1.62), pp. 27–28], which in the tilted run is a matrix in the turned directions. For p' in the dual of the unit lattice let

$$\phi_{\mu\mu'}(p') := \sum_l \frac{|u_k(p'+l)|^2}{\Delta(p'+l)} \sum_\nu V_{\mu\nu}(p'+l)\,\overline{V_{\mu'\nu}(p'+l)}\,, \tag{7.9}$$

the sum over the classes of $l \in 2\pi\mathbb{Z}^4$ modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$. By Lemma 7.1(i) and (ii) this is the momentum representation of ϕ: $(\widetilde{\phi\omega})_\mu(p') = \sum_{\mu'} \phi_{\mu\mu'}(p')\tilde\omega_{\mu'}(p')$. It is diagonal for $\mu \in \{0,1\}$, where it is the function ϕ_μ of [2, (1.62)] with the tilted u_k, and a Hermitian matrix on the indices $\{2,3\}$.

Lemma 7.3 (The Matrix ϕ)**.** *Let $32\eta \le \frac12$. There are constants $0 < \gamma_0 \le \gamma_1$, which do not depend on k and L, such that $\gamma_0|w|^2 \le \Delta_0(p')\sum_{\mu,\mu'} \bar w_\mu \phi_{\mu\mu'}(p') w_{\mu'} \le \gamma_1|w|^2$ for every $p' \in [-\pi,\pi]^4$, $p' \ne 0$, and every $w \in \mathbb{C}^4$.*

Proof. The quadratic form is $\sum_l |u_k(p'+l)|^2 \Delta_0(p')\Delta(p'+l)^{-1}|V(p'+l)^* w|^2$, a sum of non-negative terms. For the lower bound we keep the term $l = 0$. Since $|e^{it} - 1| \le |t|$, we have $\Delta(p') \le \sum_\nu |p' \cdot e'_\nu|^2 = |p'|^2$, and $\Delta_0(p') \ge (4/\pi^2)|p'|^2$ because $|e^{it} - 1| \ge 2|t|/\pi$ for $|t| \le \pi$. With Lemma 7.2(iii), applied to V^{pl} and to the adjoint of this invertible matrix, and with $|D^{\eta,1}_\mu(p')| \ge 2/\pi$ for $\mu \in \{0,1\}$, the term is at least $(4/\pi^2)(2/\pi)^8(1-10\pi\eta)^2(2/\pi)^2(1-32\eta)^2|w|^2$, and $10\pi\eta \le 32\eta \le \frac12$.

For the upper bound, $|V(p)^* w| \le \frac{8}{5}|w|$ by Lemma 7.2(ii). Let $\hat p$ be the representative of $p'+l$ with $|\eta\hat p \cdot e'_\nu| \le \pi$ for every ν. Then $\Delta(p'+l) = \Delta(\hat p) \ge (4/\pi^2)|\hat p|^2$, and $\Delta_0(p') \le |p'|^2$. For $l=0$ we have $\hat p = p'$ and the term is at most $(\pi^2/4)(8/5)^2|w|^2$. For $l \ne 0$ the difference $\hat p - p'$ is a nonzero element of $2\pi\mathbb{Z}^4$ by Lemma 7.1(iv), so that one coordinate of $\hat p$ is at least π in modulus, and $\Delta_0(p')/\Delta(p'+l) \le (\pi^2/4)|p'|^2/\pi^2 \le \pi^2$. By (7.7), $|u_k(p'+l)| \le \prod_\mu \min(1, \pi/|q_\mu|)$, since $|\partial^1_\mu| \le 2$, and each value of q, which is of the form $q_\mu = p'_\mu + 2\pi n_\mu$ with integers n_μ, belongs to at most 25 classes of l. Hence the terms with $l \ne 0$ add up to at most $25\pi^2(8/5)^2 \prod_\mu \sum_{n\in\mathbb{Z}} \min(1, \pi/|p'_\mu + 2\pi n|)^2 |w|^2 \le 25\pi^2(8/5)^2(1+\pi^2/4)^4|w|^2$, because $|p'_\mu + 2\pi n| \ge \pi(2|n|-1)$ for $n \ne 0$ and $\sum_{n\ge1}(2n-1)^{-2} = \pi^2/8$. *q.e.d.*

8 The Cells

Several of Bałaban's arguments are local. They use paths of fine bonds inside a block, and they use the lower bound for the Laplace operator of a block with Neumann conditions on the functions of mean zero, which for a cube of the fine lattice is explicit [1, (2.27), p. 580]. In the tilted run a block is a cube of the coarse lattice filled with points of the turned lattice, and neither the paths nor the bound are evident. Both follow from the description of a block as a union of *cells*, the blocks $B_{5\eta}(z)$, $z \in 5\eta\mathbb{Z}^4$, of the scale 5η: by Lemma 6.2(i) the cells of a block form a cubic array with the axes e_μ, and each of them is a translate of the same set of 5^4 points, which we now describe.

Lemma 8.1 (The Cell)**.** *Let $\mathsf{R} \subset \mathbb{Z}^2$ be the set of the pairs (m,n) with $m \in \{3,4\}$ for $n=-3$, $2 \le m \le 5$ for $n=-2$, $1 \le m \le 6$ for $n=-1$, $0 \le m \le 6$ for $n=0$, $2 \le m \le 5$ for $n=1$, and $m \in \{3,4\}$ for $n=2$.*

(i) $\mathsf{C} = \{me'_2 - ne'_3 \,|\, (m,n) \in \mathsf{R}\}$*, the vectors e'_2 and e'_3 being read in the plane. Hence the cell of $z \in 5\eta\mathbb{Z}^4$ consists of the points $z + \eta(n_0 e_0 + n_1 e_1 + me'_2 - ne'_3)$ with $n_0, n_1 \in \{0,\dots,4\}$ and $(m,n) \in \mathsf{R}$.*

(ii) Every row $\{m \,|\, (m,n) \in \mathsf{R}\}$ is an interval which contains 3 and 4.

(iii) The points $z + \eta(3e'_2 + 4e'_3)$ and $z + \eta(4e'_2 - 3e'_3)$ are the corners $z + 5\eta e_2$ and $z + 5\eta e_3$ of the neighbouring cells.

Proof. (i) An element of $\omega\mathbb{Z}[i]$ is $c = \omega(m+ni) = m\omega + n\omega i$ with integers m, n, and ω and $\omega i = (-4+3i)/5$ are the vectors e'_2 and $-e'_3$ of the plane by (2.2). Its real and imaginary parts are $(3m-4n)/5$ and $(4m+3n)/5$, so that $c \in \mathsf{C}$ if and only if $0 \le 3m-4n < 25$ and $0 \le 4m+3n < 25$, that is, $\max(\frac{4}{3}n, -\frac{3}{4}n) \le m < \min(\frac{25+4n}{3}, \frac{25-3n}{4})$. For $n = -3,\dots,2$ these bounds are $\frac{9}{4} \le m < \frac{13}{3}$, $\frac{3}{2} \le m < \frac{17}{3}$, $\frac{3}{4} \le m < 7$, $0 \le m < \frac{25}{4}$, $\frac{4}{3} \le m < \frac{11}{2}$, and $\frac{8}{3} \le m < \frac{19}{4}$, which give the rows of R. For $n \ge 3$ they require $m \ge 4$ and $m < 4$, and for $n \le -4$ they require $m \ge 3$ and $m < 3$. (ii) is read off from the list, and (iii) is (2.2). *q.e.d.*

By (ii) and (iii) two neighbouring corners of cells are joined by short paths which stay in the cell of the first: for the direction e_2 three steps $\eta e'_2$ followed by four steps $\eta e'_3$, through the points $(m,n) = (1,0),(2,0),(3,0),(3,-1),(3,-2),(3,-3)$ of R, and for the direction e_3 four steps $\eta e'_2$ followed by three steps $-\eta e'_3$, through $(1,0),\dots,(4,0),(4,1),(4,2)$. We call these paths, and the straight paths of five bonds in the directions e_0 and e_1, the *connectors* from z.

Definition 8.2 (Contours in a Block)**.** Let a be a scale, $y \in a\mathbb{Z}^4$, and $x \in B_a(y)$, let $z \in 5\eta\mathbb{Z}^4$ be the corner of the cell of x, and write $x = z + \eta(n_0 e_0 + n_1 e_1 + me'_2 - ne'_3)$ as in Lemma 8.1(i).

The contour $\Gamma^a_{y,x}$ is the following path of bonds of T_η from y to x. It first goes from y to z through corners of cells, changing the coordinates in the order $\mu = 3, 2, 1, 0$, each step from a corner to the next along the connector. It then makes n_0 steps ηe_0 and n_1 steps ηe_1, and finally it goes through the points of the cell with the plane coordinates $(1, 0)$, $(2, 0)$, $(3, 0)$, from there along the column $m = 3$ to $(3, n)$, and along the row n to (m, n).

By Lemma 8.1 every site of $\Gamma^a_{y,x}$ lies in $B_a(y)$, the part of the contour after z lies in the cell of x and has at most $4 + 4 + 9$ bonds, and the contour has at most $\frac{7}{5} \cdot 4a/\eta + 17$ bonds. These contours take the place of [2, (1.7), p. 18] and of the contours $\Gamma^{(k)}_{y,x}$ of [1, (1.4), p. 572], which are required to lie in the block.

Lemma 8.3 (A Poincaré Inequality)**.** *For every scale a, every $y \in a\mathbb{Z}^4$, and every function λ on $B_a(y)$ with $\sum_{x \in B_a(y)} \lambda(x) = 0$,*

$$\sum_{x \in B_a(y)} |\lambda(x)|^2 \le C_P \Big(\frac{a}{\eta}\Big)^2 \sum_b |\lambda(b_+) - \lambda(b_-)|^2 \,, \qquad C_P \coloneqq 1550 \,, \tag{8.1}$$

the sum on the right over the bonds $b = \langle b_-, b_+ \rangle$ of T_η with both ends in $B_a(y)$.

Proof. Write $\delta\lambda(b) \coloneqq \lambda(b_+) - \lambda(b_-)$, and let $N \coloneqq a/(5\eta)$, so that $B_a(y)$ is the union of N^4 cells, whose corners form a cube of N^4 points of $5\eta\mathbb{Z}^4$. For a point x of a cell with the corner z, the last part of the contour of Definition 8.2 is a path of at most 17 bonds inside the cell, so that $|\lambda(x) - \lambda(z)|^2$ is at most 17 times the sum of $|\delta\lambda|^2$ over the bonds of the cell; summing over the 625 points,

$$\sum_x |\lambda(x) - \lambda(z)|^2 \le 625 \cdot 17 \sum_b |\delta\lambda(b)|^2 \,, \tag{8.2}$$

the sums over the points and the bonds of the cell. Let Λ be the restriction of λ to the corners, and $\bar\Lambda$ its mean. For two neighbouring corners z and $z + 5\eta e_\mu$ in $B_a(y)$ the connector has at most 7 bonds, all with both ends in $B_a(y)$, so that $|\Lambda(z + 5\eta e_\mu) - \Lambda(z)|^2$ is at most 7 times the sum of $|\delta\lambda|^2$ over its bonds, and a bond of T_η lies on at most four connectors, those from the corner of its cell. For a function f on $\{0, \dots, N-1\}$ with the mean $\bar f$ we have $\sum_i |f_i - \bar f|^2 = \frac{1}{2N} \sum_{i,j} |f_i - f_j|^2$ and $|f_i - f_j|^2 \le N \sum_m |f_{m+1} - f_m|^2$, hence $\sum_i |f_i - \bar f|^2 \le \frac{1}{2} N^2 \sum_m |f_{m+1} - f_m|^2$. On the cube of the corners, let E_μ be the average over the coordinate μ; then $\Lambda - \bar\Lambda = \sum_{\mu=0}^3 E_0 \cdots E_{\mu-1}(1 - E_\mu)\Lambda$ is a sum of mutually orthogonal functions, the operators E_μ are contractions, and the inequality in one variable applies to $(1 - E_\mu)\Lambda$ on every line. Therefore

$$\sum_z |\Lambda(z) - \bar\Lambda|^2 \le \frac{N^2}{2} \sum_\mu \sum_z |\Lambda(z + 5\eta e_\mu) - \Lambda(z)|^2 \le \frac{N^2}{2} \cdot 7 \cdot 4 \sum_b |\delta\lambda(b)|^2 \,, \tag{8.3}$$

the last sum over the bonds with both ends in $B_a(y)$. Since λ has the mean zero, $\sum_x |\lambda(x)|^2 \le \sum_x |\lambda(x) - \bar\Lambda|^2$, and $|\lambda(x) - \bar\Lambda|^2 \le 2|\lambda(x) - \lambda(z)|^2 + 2|\Lambda(z) - \bar\Lambda|^2$ for x in the cell of z. With (8.2) and (8.3) the left side of (8.1) is at most $(2 \cdot 625 \cdot 17 + 2 \cdot 625 \cdot 14N^2) \sum_b |\delta\lambda(b)|^2 \le 38750\, N^2 \sum_b |\delta\lambda(b)|^2$, and $38750/25 = 1550$. *q.e.d.*

In the notation of [1, (2.27)], the lemma for $a = 1$ says that the Laplace operator of a unit block with Neumann conditions is at least C_P^{-1} on the functions of mean zero, where for a cube of $\eta\mathbb{Z}^4$ the bound is π^2.

9 Bounds in Position Space

Bałaban obtains the bounds of his Proposition 1.1 and the inequality [2, (1.67)] by inspection of the momentum representation, in which every operator is a scalar in the index of the field. With the matrices of Section 7 this inspection becomes laborious, and we prove the same statements in another way: four inequalities for the tilted average, proved by counting paths, the Poincaré inequality of Lemma 8.3, and the momentum representation of two scalar operators only. In this section the fine lattice is the torus T_η/Γ' with a group of periods $\Gamma' \subset \mathbb{Z}^4$, the scalar products are the sums over the points, the positive bonds, or the plaquettes, with the weight η^4 on the fine lattice and 1 on the unit lattice, and $\|\cdot\|$ are the corresponding norms. For a field A and a positive direction ν we write $(\partial'_\nu A)(b) \coloneqq \eta^{-1}(A(b+\eta e'_\nu) - A(b))$, so that $\langle A, \Delta A\rangle = \sum_\nu \|\partial'_\nu A\|^2 = \|\partial A\|^2 + \|\partial^* A\|^2$ by [2, (1.21)], and ∂_1, ∂_1^* are the operations of the unit lattice.

Lemma 9.1 (Four Inequalities for the Average)**.** *For every real field A on T_η/Γ',*

(i) $\|Q_k A\|^2 \le 2\|A\|^2$;

(ii) $\|\partial_1^* Q_k A\|^2 \le 62\,\langle A, \Delta A\rangle$;

(iii) $\|\partial_1 Q_k A\|^2 \le 24\|\partial A\|^2$;

(iv) $\|A\|^2 \le (2C_P + 64)\langle A, \Delta A\rangle + 4\|Q_k A\|^2$.

Proof. Let n_μ be the number of the bonds of a staircase σ^1_μ, so that $n_\mu \le \frac{7}{5\eta}$, and recall from the proof of Lemma 6.4(ii) that a bond of T_η lies on at most $\frac{7}{5\eta}$ of the staircases $\sigma^1_\mu(x)$, $x \in T_\eta$, $\mu = 0, \dots, 3$. We use the Cauchy–Schwarz inequality in the form $|\sum_{t\in\mathcal{T}} c_t|^2 \le |\mathcal{T}| \sum_t |c_t|^2$.

(i) By (6.3), $(Q_k A)(\langle y, y + e_\mu\rangle)$ is η^5 times a sum of the values of A over the $\eta^{-4} n_\mu$ pairs of a point $x \in B_1(y)$ and a bond of $\sigma^1_\mu(x)$. Its square is therefore at most $\eta^{10}\eta^{-4} n_\mu$ times the sum of $|A(b)|^2$ over these pairs, and in the sum over y and μ every bond occurs at most $\frac{7}{5\eta}$ times. Hence $\|Q_k A\|^2 \le \frac{49}{25}\eta^4 \sum_b |A(b)|^2$.

(ii) We have $(\partial_1^* B)(y) = \sum_\mu (B(\langle y - e_\mu, y\rangle) - B(\langle y, y + e_\mu\rangle))$, and $(Q_k A)(\langle y, y + e_\mu\rangle) - (Q_k A)(\langle y - e_\mu, y\rangle) = \eta^5 \sum_{x\in B_1(y)} (A(\sigma^1_\mu(x)) - A(\sigma^1_\mu(x - e_\mu)))$. The staircase $\sigma^1_\mu(x)$ is the translate of $\sigma^1_\mu(x - e_\mu)$ by e_μ. For a bond b the difference $A(b + e_\mu) - A(b)$ is the sum of the n_μ differences of A between the translates of b by two consecutive sites of $\sigma^1_\mu(0)$, each of which is $\pm\eta(\partial'_\nu A)(b'')$ for a translate b'' of b and a direction ν. The difference of the two averages is therefore $\eta^5 \cdot \eta$ times a sum of $\eta^{-4} n_\mu^2$ values $\pm(\partial'_\nu A)(b'')$, and in the sum over y every pair (b'', ν) occurs at most n_μ^2 times. This gives $\sum_y |\cdot|^2 \le \eta^{12}\eta^{-4} n_\mu^4 \sum_{b'',\nu} |(\partial'_\nu A)(b'')|^2 \le (\frac{7}{5})^4 \langle A, \Delta A\rangle$ for every μ, and $|(\partial_1^* Q_k A)(y)|^2$ is at most four times the sum of the squares of the four differences; $16(\frac{7}{5})^4 \le 62$.

(iii) Let P be the plaquette of the unit lattice with the corners y, $y + e_\mu$, $y + e_\mu + e_\nu$, $y + e_\nu$. Since $B_1(y + e_\mu) = B_1(y) + e_\mu$, we have $(\partial_1 Q_k A)(P) = \eta^5 \sum_{x\in B_1(y)} A(\ell(x))$ with the closed path $\ell(x)$ which runs along $\sigma^1_\mu(x)$ and $\sigma^1_\nu(x + e_\mu)$ and back along $\sigma^1_\mu(x + e_\nu)$ and $\sigma^1_\nu(x)$. Let $s_0 = 0, s_1, \dots, s_{n_\mu} = e_\mu$ be the sites of $\sigma^1_\mu(0)$ and $s'_0, \dots, s'_{n_\nu}$ those of $\sigma^1_\nu(0)$. The points $x + s_i + s'_j$ form a mesh whose boundary is $\ell(x)$, and the sum of A over $\ell(x)$ is the sum over i and j of the sums of A over the closed paths through $x + s_i + s'_j$, $x + s_{i+1} + s'_j$, $x + s_{i+1} + s'_{j+1}$, $x + s_i + s'_{j+1}$. If the two steps $s_{i+1} - s_i$ and $s'_{j+1} - s'_j$ are parallel, such a path runs through each of its bonds once in each sense and contributes nothing; otherwise it is the boundary of a plaquette p of T_η and contributes $\pm\eta(\partial A)(p)$. Hence $(\partial_1 Q_k A)(P)$ is $\eta^5 \cdot \eta$ times a sum

of at most $\eta^{-4} n_\mu n_\nu$ values $\pm(\partial A)(p)$, and in the sum over y every plaquette occurs at most $n_\mu n_\nu$ times. This gives $\sum_y |(\partial_1 Q_k A)(P)|^2 \le (\frac{7}{5})^4 \|\partial A\|^2$ for each of the six pairs $\mu < \nu$, and $6(\frac{7}{5})^4 \le 24$.

(iv) Let $A_\nu(x) \coloneqq A(\langle x, x+\eta e'_\nu\rangle)$, and let $\bar{A}_\nu(y)$ be the mean of A_ν over $B_1(y)$. Lemma 8.3 with $a = 1$, applied to $A_\nu - \bar{A}_\nu(y)$ on every block, gives $\sum_\nu \sum_x \eta^4 |A_\nu(x) - \bar{A}_\nu(y(x))|^2 \le C_P \langle A, \Delta A\rangle$, where $y(x)$ is the corner of the block of x, because the differences of A_ν along the bonds of a block are among the numbers $\eta(\partial'_{\nu'} A)(b)$. Since a block has η^{-4} points, $\|A\|^2 \le 2C_P \langle A, \Delta A\rangle + 2\sum_y \sum_\nu |\bar{A}_\nu(y)|^2$. By Lemma 6.4(ii) the average of the constant field with the values $\bar{A}_\nu(y)$ is $\sum_\nu (e_\mu \cdot e'_\nu)\bar{A}_\nu(y)$, so that $(Q_k A)(\langle y, y+e_\mu\rangle) - \sum_\nu (e_\mu \cdot e'_\nu)\bar{A}_\nu(y)$ is η^5 times the sum, over $x \in B_1(y)$ and over the bonds of $\sigma^1_\mu(x)$, of the differences between the value of A on the bond and the value of the same component at x. Each of these differences is a sum of at most n_μ terms $\pm\eta(\partial'_{\nu'} A)(b'')$ along the staircase, and the counting of (ii) bounds the sum over y of the squares by $(\frac{7}{5})^4 \langle A, \Delta A\rangle$ for every μ. The matrix $(e_\mu \cdot e'_\nu)$ is orthogonal, so that $\sum_\nu |\bar{A}_\nu(y)|^2 \le 2\sum_\mu |(Q_k A)(\langle y, y+e_\mu\rangle)|^2 + 2\sum_\mu |\cdot|^2$ with the differences just bounded, and $\sum_y \sum_\nu |\bar{A}_\nu(y)|^2 \le 2\|Q_k A\|^2 + 8(\frac{7}{5})^4 \langle A, \Delta A\rangle$. Since $16(\frac{7}{5})^4 \le 64$, this is (iv). *q.e.d.*

The two scalar operators are those of [2, (1.38), (1.58)] for functions. As there, Δ^{-1} denotes the inverse of the Laplace operator $\Delta = \partial^* \partial$ of T_η/Γ' on the functions orthogonal to the constants, and $\Delta_0 \coloneqq \partial_1^* \partial_1$ is the Laplace operator of the unit lattice.

Lemma 9.2 (Two Scalar Operators)**.** *Let $10\pi\eta \le \frac{1}{2}$. On the functions of the unit lattice which are orthogonal to the constants, the operators $\mathsf{S} \coloneqq Q'_k \Delta^{-2} Q'^*_k$ and $\mathsf{T} \coloneqq Q'_k \Delta^{-1} Q'^*_k$ satisfy*

$$\mathsf{S} \ge c_{\mathsf{S}} \Delta_0^{-2} , \qquad \mathsf{T} \le C_{\mathsf{T}} \Delta_0^{-1} , \qquad c_{\mathsf{S}} \coloneqq \tfrac{1}{4}(\tfrac{2}{\pi})^{12} , \quad C_{\mathsf{T}} \coloneqq \tfrac{\pi^2}{4} + 100(1 + \tfrac{\pi^2}{4})^4 . \tag{9.1}$$

Proof. By Lemma 7.1(i) and its adjoint, both operators are multiplications in the momentum representation of the unit lattice, by $\mathsf{S}(p') = \sum_l |u_k(p'+l)|^2 \Delta(p'+l)^{-2}$ and $\mathsf{T}(p') = \sum_l |u_k(p'+l)|^2 \Delta(p'+l)^{-1}$, for $p' \in [-\pi, \pi]^4$, $p' \ne 0$. As in the proof of Lemma 7.3, $(4/\pi^2)|p'|^2 \le \Delta_0(p') \le |p'|^2$ and $(4/\pi^2)|p'|^2 \le \Delta(p') \le |p'|^2$. The term $l = 0$ of S is at least $(2/\pi)^8 (1 - 10\pi\eta)^2 |p'|^{-4}$ by (7.8), and $|p'|^{-4} \ge (4/\pi^2)^2 \Delta_0(p')^{-2}$. The term $l = 0$ of T is at most $(\pi^2/4)|p'|^{-2} \le (\pi^2/4)\Delta_0(p')^{-1}$. For $l \ne 0$ we have $\Delta(p'+l) \ge 4$ and $\sum_{l \ne 0} |u_k(p'+l)|^2 \le 25(1 + \pi^2/4)^4$ by the proof of Lemma 7.3, and $1 \le 16\Delta_0(p')^{-1}$. *q.e.d.*

Proposition 9.3 (Proposition 1.1 and the Inequality (1.67) for the Tilted Run)**.** *Let $32\eta \le \frac{1}{2}$ and $a > 0$. There are constants $\gamma_0 > 0$ and C, which depend on a and do not depend on k, L, and the torus, such that the operator $\Delta_a = \Delta - \partial P \partial^* + a Q_k^* Q_k$ of [2, (1.69)] and its inverse G satisfy*

$$\Delta_a \ge \gamma_0 (\Delta + I) , \tag{9.2}$$

$$\|GJ\|, \ \|\nabla GJ\|, \ \|G\nabla^* J\|, \ \|\nabla G \nabla^* J\| \le \gamma_0^{-1} \|J\| , \qquad \|\nabla\nabla GJ\|, \ \|G\nabla^*\nabla^* J\| \le C\|J\| , \tag{9.3}$$

where $\nabla A \coloneqq (\partial'_\nu A)_\nu$, and such that the action Δ_k of [2, (1.65)] satisfies

$$\tfrac{1}{24} \|\partial_1 B\|^2 \le \langle B, \Delta_k B\rangle \le \gamma_\phi^{-1} \|\partial_1 B\|^2 , \tag{9.4}$$

where γ_ϕ denotes the lower constant γ_0 of Lemma 7.3.

Proof. We use the operator identities of [2], which hold in the tilted run by Section 10: $R = I - P$ is the orthogonal projection onto $\Delta N(Q'_k)$, where $N(Q'_k)$ is the null space of

Q'_k; $P = \Delta^{-1} Q'^*_k \mathsf{S}^{-1} Q'_k \Delta^{-1}$ on the functions orthogonal to the constants; and $\langle A, \Delta_a A\rangle = \|\partial A\|^2 + \|R\partial^* A\|^2 + a\|Q_k A\|^2$.

A gauge transformation. Let A be a field. There is $\lambda \in N(Q'_k)$ with $\Delta\lambda = R\partial^* A$, and $A_s := A - \partial\lambda$ satisfies $\partial A_s = \partial A$, $Q_k A_s = Q_k A - \partial_1 Q'_k \lambda = Q_k A$ by Lemma 6.4(i), and $\partial^* A_s = \partial^* A - \Delta\lambda = P\partial^* A$. Hence

$$\langle A, \Delta_a A\rangle = \|\partial A_s\|^2 + a\|Q_k A_s\|^2 + \|\Delta\lambda\|^2 \,. \tag{9.5}$$

The gauge part. Since $\partial\partial\lambda = 0$, we have $\langle \partial\lambda, \Delta\partial\lambda\rangle = \|\partial^*\partial\lambda\|^2 = \|\Delta\lambda\|^2$. The function λ has the mean zero on every unit block, so that $\|\lambda\|^2 \le C_P\|\partial\lambda\|^2$ by Lemma 8.3 with $a = 1$, and $\|\partial\lambda\|^2 = \langle\lambda, \Delta\lambda\rangle \le \|\lambda\|\|\Delta\lambda\|$; together, $\|\partial\lambda\|^2 \le C_P\|\Delta\lambda\|^2$.

The divergence on the slice. Put $f := \partial^* A_s$, so that $f = Pf$ is orthogonal to the constants, and $\mu := Q'_k \Delta^{-1} f$. Then $\|f\|^2 = \langle f, Pf\rangle = \langle \mu, \mathsf{S}^{-1}\mu\rangle \le c_{\mathsf{S}}^{-1}\|\Delta_0\mu\|^2$ by Lemma 9.2. The field $A^T := A_s - \partial\Delta^{-1} f$ satisfies $\partial^* A^T = 0$ and $\partial A^T = \partial A_s$, and $\partial_1\mu = Q_k\partial\Delta^{-1} f = Q_k A_s - Q_k A^T$ by Lemma 6.4(i). Hence $\Delta_0\mu = \partial_1^* Q_k A_s - \partial_1^* Q_k A^T$, where $\|\partial_1^* Q_k A_s\|^2 \le 16\|Q_k A_s\|^2$, each bond of the unit lattice entering ∂_1^* at its two ends, and $\|\partial_1^* Q_k A^T\|^2 \le 62\langle A^T, \Delta A^T\rangle = 62\|\partial A_s\|^2$ by Lemma 9.1(ii). Therefore

$$\|\partial^* A_s\|^2 \le C_4(\|\partial A_s\|^2 + \|Q_k A_s\|^2) \,, \qquad C_4 := 124\, c_{\mathsf{S}}^{-1} \,. \tag{9.6}$$

The lower bound. By (9.6), $\langle A_s, \Delta A_s\rangle \le (1 + C_4)(\|\partial A_s\|^2 + \|Q_k A_s\|^2)$, and by Lemma 9.1(iv), $\|A_s\|^2 \le (2C_P + 64)\langle A_s, \Delta A_s\rangle + 4\|Q_k A_s\|^2$. With the bounds on the gauge part and $\langle A, (\Delta + I)A\rangle \le 2\langle A_s, (\Delta + I)A_s\rangle + 2\langle\partial\lambda, (\Delta + I)\partial\lambda\rangle$, the identity (9.5) gives (9.2) with

$$\gamma_0^{-1} := 2\max\Big(\frac{(2C_P + 65)(1 + C_4) + 4}{\min(1, a)},\ 1 + C_P\Big) \,. \tag{9.7}$$

The bounds on G. By (9.2), Δ_a is invertible and $(\Delta + I)^{1/2} G (\Delta + I)^{1/2}$ has the norm at most γ_0^{-1}; since $\|\nabla A\|^2 = \langle A, \Delta A\rangle$, this is the first half of (9.3). On the torus the operators ∂'_ν commute with each other and with their adjoints, so that $\|\nabla\nabla A\|^2 = \|\Delta A\|^2$, and $\Delta G = I + \partial P\partial^* G - aQ_k^* Q_k G$. Here $\|Q_k^* Q_k G J\| \le 2\gamma_0^{-1}\|J\|$ by Lemma 9.1(i). For a function g orthogonal to the constants, $Pg = \Delta^{-1} Q'^*_k \omega$ with $\omega = \mathsf{S}^{-1} Q'_k \Delta^{-1} g$, so that $\|\partial Pg\|^2 = \langle Pg, \Delta Pg\rangle = \langle\omega, \mathsf{T}\omega\rangle \le C_{\mathsf{T}}\langle\omega, \Delta_0^{-1}\omega\rangle$, while $\|g\|^2 \ge \|Pg\|^2 = \langle\omega, \mathsf{S}\omega\rangle \ge c_{\mathsf{S}}\langle\omega, \Delta_0^{-2}\omega\rangle \ge \frac{1}{16} c_{\mathsf{S}}\langle\omega, \Delta_0^{-1}\omega\rangle$, because $\Delta_0 \le 16$. With $g = \partial^* GJ$, $\|g\| \le \|\nabla GJ\|$, this gives $\|\Delta GJ\| \le (1 + 2a\gamma_0^{-1} + (16C_{\mathsf{T}}/c_{\mathsf{S}})^{1/2}\gamma_0^{-1})\|J\|$, and the last bound of (9.3) follows by taking adjoints.

The action. By [2, (1.47)–(1.65)], $\langle B, \Delta_k B\rangle = \|\partial H_k B\|^2$ is the minimum of $\|\partial A\|^2$ over the fields with $Q_k A = B$ and $R\partial^* A = 0$, and by the gauge transformation above the second condition may be dropped. Lemma 9.1(iii) for $A = H_k B$ is the lower bound of (9.4). For the upper bound write $B = B_0 + B'$ with B_0 constant and B' orthogonal to the constants, and $B' = B^T + \partial_1\nu$ with $\nu := \Delta_0^{-1}\partial_1^* B'$, so that $\partial_1^* B^T = 0$ and $\langle B^T, \Delta_0 B^T\rangle = \|\partial_1 B^T\|^2 = \|\partial_1 B\|^2$. Let $\phi = Q_k\Delta^{-1} Q_k^*$, whose momentum representation is (7.9); by Lemma 7.3, $\phi^{-1} \le \gamma_\phi^{-1}\Delta_0$ on the fields orthogonal to the constants. The field $A := Q_k^* B_0 + \Delta^{-1} Q_k^*\phi^{-1} B^T + \partial Q'^*_k\nu$ satisfies $Q_k A = B_0 + B^T + \partial_1 Q'_k Q'^*_k\nu = B$, by Lemma 6.4 and because $Q'_k Q'^*_k = I$, and $\partial A = \partial\Delta^{-1} Q_k^*\phi^{-1} B^T$. Hence $\langle B, \Delta_k B\rangle \le \|\partial A\|^2 \le \langle\Delta^{-1} Q_k^*\phi^{-1} B^T, Q_k^*\phi^{-1} B^T\rangle = \langle B^T, \phi^{-1} B^T\rangle \le \gamma_\phi^{-1}\|\partial_1 B\|^2$. *q.e.d.*

The proposition has the content of [2, Proposition 1.1, (1.89), (1.90), and (1.67)]; the two bounds on second derivatives hold with a constant other than γ_0^{-1}.

10 The Modifications in the First Paper on the Propagators

We go through [2] in its order. The paper treats the Gaussian theory of one real field, with the action $\frac{1}{2}\langle \partial A, \partial A\rangle$ on the fine lattice, and it derives the propagators of the later papers from the averages. In the tilted run its fine lattice is T_η of (6.1), on a torus T_η/Γ' with a group of periods $\Gamma' \subset \mathbb{Z}^4$ in place of the box of [2, p. 17]; the operators ∂, ∂^*, and Δ are those of this lattice, whose bonds have the directions e'_ν, and a field has the components A_ν in these directions, while the fields B and the multipliers ω on the unit lattice have the components in the directions e_μ.

Section A, (1.1)–(1.15). The action [2, (1.3), (1.5)] and its gauge invariance [2, (1.4)] concern the fine lattice alone and are unchanged. The blocks [2, (1.6)] are those of Definition 6.1(i), and the average [2, (1.8), (1.11)] is replaced by Definition 6.1(iv); the transformation laws [2, (1.9), (1.13)] are Lemma 6.4(i). The contours [2, (1.7)] enter the axial gauge condition [2, (1.10)], which requires for every point x of a block a path of fine bonds from the corner y to x; we take the path which changes the coordinates of $x - y$ in the basis $\eta e'_\nu$ in the order $\nu = 3, 2, 1, 0$. Such a path may leave the block, by less than its side, which is of no consequence in [2, (1.10)–(1.15)]: these use only that the conditions $A(\Gamma_{y,x}) = 0$ fix the gauge transformations which vanish at the corners, one condition for each point $x \neq y$.

Section B, (1.16)–(1.20). The computation [2, (1.16)] of a composition is Lemma 6.3: the average of the first step of the tilted run is $Q_{L^{-1}}$ in the unit of the lattice reached after k steps, the later ones are Bałaban's, and the composition is Q_k. The formula [2, (1.20)] is Lemma 6.4(i).

Sections C to E, the operator identities. The derivations of [2, (1.21)–(1.28), (1.38)–(1.44), (1.46)–(1.60), (1.64), (1.65), (1.68)–(1.82), (1.91)–(1.107)] are calculations with the operators ∂, ∂^*, Δ of the fine lattice, with Q_k, Q'_k, their adjoints, and with ∂_1 on the unit lattice. They use the following properties and no others: the decomposition [2, (1.21)] of the action, which holds on every lattice with four orthogonal directions of bonds and hence on T_η; the identity [2, (1.55)], which is Lemma 6.4(i); that Q'_k maps constants to constants and their orthogonal complement to the orthogonal complement; and, in [2, (1.57)–(1.59), (1.74), (1.82)], that Q_k and Q_k^* map the constant fields to the constant fields with $Q_k^* Q_k = 1$ on them. The last two properties are Lemma 6.4(ii). These derivations therefore hold without change, and so do their results in operator form: the projection R [2, (1.38), (1.44)], the operator H_k [2, (1.60), (1.103)], the action Δ_k [2, (1.65)], the operator G [2, (1.81), (1.82)], and the identities [2, (1.95), (1.97), (1.99), (1.107)].

The momentum representation, (1.29)–(1.37) and (1.45). The Fourier transform [2, (1.29)] is that of the finite group T_η/Γ', whose dual consists of the classes modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$ of the p with $p \cdot \gamma \in 2\pi\mathbb{Z}$ for $\gamma \in \Gamma'$, and the decomposition $p = p' + l$ is as described after Lemma 7.1. The symbols [2, (1.31)] are replaced by (7.1) and (7.4). The solution [2, (1.32)–(1.35), (1.37)] uses of u_k only that $u_k(l) = 0$ for the classes $l \neq 0$ and $u_k(0) = 1$, which is Lemma 7.1(iv), and it holds without change. The bound $\Delta_0(p')/\Delta(p' + l) \leq O(1)(1 + |l|^2)^{-1}$ holds with l replaced by $\hat{l} \coloneqq \hat{p} - p'$ for the representative $\hat{p}$ of the proof of Lemma 7.3: by the estimates given there the quotient is at most $\pi^4/|\hat{p}|^2$, and $|\hat{p}| \geq \max(\pi, |\hat{l}| - 2\pi)$ for $\hat{l} \neq 0$. The bound [2, (1.36)] is replaced by (7.7). The formula [2, (1.45)] and the bounds on it are taken from [1], which we have not examined.

The momentum representation, (1.61)–(1.63), (1.66), (1.67), (1.83)–(1.88), and (1.101). The representation [2, (1.61)] of Q_k is replaced by Lemma 7.1(ii): the scalar v_μ becomes the matrix V. The function ϕ_μ of [2, (1.62)] becomes the matrix (7.9), and the bounds

$\gamma_0 \le \Delta_0 \phi_\mu \le \gamma_1$ stated there are Lemma 7.3. The formulae [2, (1.63), (1.66), (1.83)–(1.88), (1.101)] are the operator formulae [2, (1.60), (1.65), (1.81), (1.99)] written in components under the assumption that ϕ is diagonal. In the tilted run they have to be written with the matrices V and ϕ on the indices $\{2,3\}$, in which products such as $\overline{v_\mu}\phi_\mu^{-1}$ become $V^*\phi^{-1}$ and the function ϕ_μ of [2, (1.84)] becomes $\mathbf{1} + a\phi$. Bałaban reads three bounds off from these formulae. Two of them, the inequality [2, (1.67)] and Proposition 1.1 [2, (1.89), (1.90)], are Proposition 9.3, which we have proved without the formulae in components. The third, the bound on the sum over l stated after [2, (1.63)], from which the Hölder continuity of the derivatives of $H_k B$ follows, has not been carried out; it belongs with Section F.

Section F, Proposition 1.2. Not carried out. Its proof has three parts. The first is the expansion [2, (1.119)–(1.123)] of G in the operators $h_z G h_z$ of a partition of unity over cubes of the size M_0 and in the commutators [2, (1.120), (1.121)] of the averages with the multiplication by a function, in which the segments $[x, x^{(k)}]$ are to be replaced by the staircases and the contours by those of Definition 8.2; the L^2-bounds which it needs are (9.3). The second is the pair of bounds [2, (1.126), (1.127)] on the kernel of $\partial P \partial^*$, and the third the reduction [2, (1.132)–(1.137)] of the bounds in the supremum and Hölder norms to the scalar operator G'. Both rest on [1, Lemma 2.4 and (2.34)], for which see Section 11.

11 The Modifications in the Paper on the Scalar Green's Functions

The paper [1] treats the operators $G_k(\Omega, A) = (-\Delta^\eta_{A,\Omega} + m^2 + aP_k(A))^{-1}$ for functions with values in $\mathbb{R}^N$, on a union Ω of big blocks, with Neumann conditions on its boundary and with an external field A. We have read its Sections 1 and 2, with the following result for the tilted run.

The setting, (1.1)–(1.8). The blocks [1, (1.1)] are those of Definition 6.1(i), and the big blocks are cubes of the unit lattice as before. The quadratic form [1, (1.3)] is the sum over the bonds of T_η with both ends in Ω. The covariant average [1, (1.4)] needs contours inside the blocks, which are those of Definition 8.2.

The random walk representation, (2.2)–(2.13). The partition of unity consists of functions of the coordinates x_μ/M, whose differences along a bond of T_η are $O(\eta/M)$ as before. The formulae [1, (2.3)–(2.5)] concern one bond or the bonds at one point and do not involve the blocks; the commutator [1, (2.8)] of $P_k(A)$ with a function uses the contours of the block only. The derivation of [1, (2.9)–(2.13)] is therefore unchanged.

Lemma 2.1 and Corollary 2.3. The proof of the lemma reduces the L^2-bounds, by the expansion [1, (2.24)–(2.26)] in the regular part of the field, to the lower bound [1, (2.27)] for one unit block at $A = 0$, where the Laplace operator of the block with Neumann conditions is bounded below by π^2 on the functions orthogonal to the constants. In the tilted run this bound is Lemma 8.3, with C_P^{-1} in place of π^2, so that [1, (2.28)] holds with $c_0^{-1} = \min\{C_P^{-1}, a_k\}$, and [1, (2.29)] is unchanged. We have not checked the expansion [1, (2.24)] itself, which is quoted from another paper. Corollary 2.3 follows from the lemma and the random walk representation as in the text.

Lemma 2.2 and the last assertion of the Theorem. These concern a rectangular parallelepiped $\square$ with Neumann conditions, and their proof passes from the operator of the whole lattice to that of $\square$ by multiple reflections [1, (2.42)]. The reflections in the faces of $\square$ which are orthogonal to e_2 or e_3 do not map T_η to itself: by Lemma 5.1 no reflection of the plane of these two directions preserves both lattices. The formula [1, (2.42)] is therefore not available in the tilted run, and neither are, by this proof, the bounds of Lemma 2.2 in the supremum,

Hölder, and L^p norms up to the boundary of a parallelepiped. In the proof of the Theorem the lemma is applied to the cubes $\square_j$ in the interior of Ω, on which the functions h_j vanish near the boundary; a replacement of the operators of these cubes by operators without a boundary has not been carried out.

Lemma 2.4. For the whole lattice the representation [1, (2.43)–(2.48)] is the solution of a linear equation in the momentum representation, and it holds with the function u_j of (7.4). The bounds [1, (2.49)–(2.51)] and the analytic continuation in p' on which the exponential decay rests have not been carried out; the upper bound on u_j used there is replaced by (7.7), and of the lower bound stated there for all l, which we have not proved for the tilted u_j, only the case $l = 0$ enters [1, (2.50)], and this case is (7.8). The identity [1, (2.34)] is quoted from a paper which is not among our sources.

Sections 3 to 5. Not examined.

12 What Is Not Covered

The equality (1.1) is not proved. By Proposition 4.2 it is equivalent to a comparison of the construction of the paper on Λ_K, on the torus $\mathbb{T}_\Gamma$, with the tilted run, and for this comparison the following has not been examined.

In [2]: the bound after its (1.63) and its Section F, as described in Section 10.

In [1]: the bounds in the supremum, Hölder, and L^p norms, for which the proof by reflections is not available and no replacement has been carried out; the bounds and the analytic continuation in the proof of its Lemma 2.4; its identity (2.34); and its Sections 3 to 5.

The other papers on the linear theory. The averaging operations for fields with values in G [4], and [3, 6, 5, 7]. Every background field, propagator, and gauge fixing of the tilted run, at every step, is defined on the turned fine lattice under the constraints of the averages of Section 6.

The coupling. The consequences of the modification stated at the end of Section 5 for the description of the densities after k steps [9, Theorem 1, p. 262] and for the choice of the bare coupling (Paper, Theorem 4.1).

The paper. Its construction on the torus $\mathbb{T}_\Gamma$, whose periods are not multiples of the vectors of the basis; the first step with a loop ensemble (Paper, Theorem 5.2) for the tilted run; and the comparison of two runs, in which the identities of Paper, Lemma 9.2, use of the passage between the two fine lattices only a linear map with a right inverse, while the bounds on the defect (Paper, Proposition 9.3), on the minimisers, and on the large-field terms are proved for two fine lattices of which one is the lattice of the cubes of the other. The normalisation of the tilted run is a product of its own step factors; since ρ fixes a loop in the plane $\{z = 0\}$, the comparison for a single loop relates it to that of the paper whenever the limit for this loop is not zero.

References

[1] Tadeusz Bałaban, *Regularity and Decay of Lattice Green's Functions*, Comm. Math. Phys. **89** (1983), 571–597.

[2] ———, *Propagators and Renormalization Transformations for Lattice Gauge Theories. I*, Comm. Math. Phys. **95** (1984), 17–40.

[3] ———, *Propagators and Renormalization Transformations for Lattice Gauge Theories. II*, Comm. Math. Phys. **96** (1984), 223–250.

[4] ———, *Averaging Operations for Lattice Gauge Theories*, Comm. Math. Phys. **98** (1985), 17–51.

[5] ———, *Propagators for Lattice Gauge Theories in a Background Field*, Comm. Math. Phys. **99** (1985), 389–434.

[6] ———, *Spaces of Regular Gauge Field Configurations on a Lattice and Gauge Fixing Conditions*, Comm. Math. Phys. **99** (1985), 75–102.

[7] ———, *The Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, Comm. Math. Phys. **102** (1985), 277–309.

[8] ———, *Renormalization Group Approach to Lattice Gauge Field Theories. I. Generation of Effective Actions in a Small Field Approximation and a Coupling Constant Renormalization in Four Dimensions*, Comm. Math. Phys. **109** (1987), 249–301.

[9] ———, *Convergent Renormalization Expansions for Lattice Gauge Theories*, Comm. Math. Phys. **119** (1988), 243–285.